

Solutions to Five Challenge Problems in Enumerative and Algorithmic Combinatorics, with an Account of the Human–Machine Methodology Employed

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Abstract

We report solutions to five challenge problems posed by Doron Zeilberger and his collaborators, together with substantial partial progress on two further problems, and we describe the method by which they were obtained.

The solved problems are: the Second Computational Chomp Challenge of Ekhad and Zeilberger, for which we exhibit a bar with three winning opening moves; the third challenge of Spahn and Zeilberger, asking whether the restricted permutation counts $a_{r,s}$ and $b_{r,s}$ are holonomic for all $r, s > 1$, answered affirmatively; the First Rigorous Solid Standard Young Tableaux Challenge, for which we prove the conjectured second-order recurrence; the five-dimensional Geode Challenge of Amdeberhan, Kauers and Zeilberger; and Conjectures 2a and 2b of Kauers and Zeilberger, which we obtain from a local limit theorem for excursions of Markov-modulated random walks in cones.

Several results of independent interest arise along the way: a staircase theorem constraining the winning opening moves of any Chomp bar, together with a parity theorem for square bars; an explicit algebraic generating function for reverse-Kreweras diagonal walks and a closed form for their diagonal-endpoint counts; a one-dimensional integral representation for diagonal Geode coefficients; and the identity that each Kauers–Zeilberger constant is a universal factor times the square of the apex value of a discrete cone-harmonic function.

All of the work reported here was carried out in collaboration with a large language model. The division of labour is described in Section 1.2; Section 2 records the verification protocol this mode of work required, and Section 11 documents the failures, which we regard as an essential part of the report.

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Part I

Introduction and method

1 Introduction

Zeilberger and his collaborators have, over several decades, posed a large number of concrete challenge problems, many carrying pledged donations to the On-Line Encyclopedia of Integer Sequences in honour of the solver. The problems are heterogeneous: some are computational records, some are requests for proofs of empirically observed recurrences, and some are genuine open problems in enumerative combinatorics.

Between June and August 2026 the author worked systematically through this collection. This paper reports the outcomes and the method.

1.1 Summary of results

Problem	Status	Section
Second Computational Chomp Challenge	solved	4
Spahn–Zeilberger Challenge 3 (holonomicity)	solved	5
First Rigorous Solid SYT Challenge	solved	7
Geode Challenge, $D = 5$ (and $D = 6, \dots, 10$)	solved	10
Kauers–Zeilberger Conjectures 2a, 2b	solved	9
Spahn–Zeilberger Challenge 1 ($a_{2,2}$ operator)	partial	6
Second Rigorous Solid SYT Challenge (cone exponent)	partial	8

1.2 Disclosure of method

The mathematics reported here was produced in extended collaboration with a large language model (Anthropic’s Claude). The division of labour was as follows. The model generated the mathematical ideas, the proofs, and the verification code. The author directed the investigation: selected which problems to attack and when to abandon them, executed all computations, designed and enforced the verification protocol of Section 2, and identified a substantial number of errors in the model’s output, a representative sample of which is catalogued in Section 11.

We state this at the outset for two reasons. First, it is materially relevant to how the results should be assessed: a reader may wish to weight machine-checked claims differently from prose arguments, and we indicate throughout which is which. Second, the practical lessons of working in this mode are, in our judgement, of comparable interest to the individual theorems, and cannot be conveyed without the disclosure.

2 Methodology

The verification protocol below emerged from failure rather than from design. Each of its components was adopted after an error that it would have prevented.

2.1 External validation precedes construction

The governing principle is that a derivation must be checked against a quantity independently known to be correct, as early and as cheaply as possible. Internal consistency — verifying a construction against itself, or against a second construction sharing its assumptions — is not validation.

The clearest instance arose in the work of Section 6. A chain of five successive reductions had each been verified against the preceding one. A twenty-line script then evaluated the composite against five exact values established by independent means, together with two further values the chain had never been shown. All seven agreed. Had they not, the error would have been localised immediately; as it was, the check converted a plausible chain into a trustworthy one at negligible cost.

2.2 New capabilities are gated on solved problems

Before applying a new computational tool to an open problem, we apply it to a problem whose answer is already established, and require exact agreement.

This detected two significant faults. In one case a computer algebra library’s multivariate creative-telescoping routine was found to fail on the library’s own documented examples — an error in the software, not in our encoding, and one that would otherwise have been attributed to the mathematics. In another, a sign error in the annihilator of an exponential factor had corrupted two days of output, and was identified only when the resulting discrepancy was recognised as exactly twice a known boundary term.

2.3 Failure modes are distinguished instrumentally

A search returning no solution is ambiguous between an ansatz that is too small and a formulation that is structurally wrong; the two require opposite responses. We instrumented our solvers to report the number of candidate solutions existing *before* the data check was applied. Zero such candidates indicates the former; candidates that exist but fail the data indicates the latter. Without this instrumentation the two are indistinguishable from outside, and we lost considerable effort enlarging ansatz spaces within a formulation that could not have contained the answer.

2.4 Standards of evidence

We distinguish throughout:

- **proved**: a complete mathematical argument is given;
- **machine-verified**: an identity has been confirmed symbolically or by exact arithmetic over a stated range;
- **supported**: a pattern has been confirmed on data but not proved, with the range stated.

Section 4.6 contains an instance in which a pattern confirmed on five consecutive values proved false at the sixth. We regard the distinction between the second and third categories as essential, and we have tried to make it visible at every claim.

3 The role of the automated system

Because the division of labour described in Section 1.2 is unfamiliar, and because it determined both the successes and the failures recorded here, we set it out in detail. The account is descriptive rather than advocatory; we report what occurred.

3.1 Division of labour

The investigation proceeded as an extended dialogue, one conversation per problem, over approximately two months. Within each conversation the roles were stable.

The automated system proposed reductions, constructed proofs, derived closed forms, and wrote every line of the verification and computation code. It also produced the first drafts of the manuscripts on which Sections 4, 5, 7, 9 and 10 are based.

The author selected the problems and determined when a line of attack should be abandoned; executed all computations, on hardware the system could not access; designed and enforced the verification protocol of Section 2; adjudicated between competing proposals; and identified errors, of which Section 11 records a representative sample.

Neither role was passive. The system did not merely formalise ideas supplied to it, and the author did not merely execute instructions. The characteristic interaction was: a proposal, an objection or a request for a cheaper check, a computation, and a revision — iterated, frequently for many cycles, before anything survived.

3.2 The characteristic failure mode

The dominant practical fact about this mode of work is that the system’s failures are not silences but fluent, plausible, incorrect output. A proposed reduction that is subtly wrong looks exactly like one that is right; a solver that returns nothing because of a floating-point defect looks exactly like one that returns nothing because no solution exists.

This has two consequences that shaped the entire project.

First, *every* substantive claim required external validation before being built upon. The protocol of Section 2 is not a matter of good practice but of necessity: without it, an incorrect intermediate result propagates silently into all subsequent work, and the eventual contradiction surfaces far from its cause. The most expensive episodes recorded in Section 11 are precisely those in which a check was deferred.

Second, the author’s principal contribution was adversarial rather than generative. The questions that repeatedly proved most productive were: against what independently known quantity has this been checked; what would the output look like if this were wrong; and how long will this computation take. The third question was, empirically, the one most often neglected by the system and most often decisive.

3.3 Loops and their termination

A recurring pathology was circular reasoning: the system would propose a reduction, encounter an obstruction, propose a second reduction that reintroduced the first obstruction in different notation, and continue indefinitely. Left unchecked these loops consumed hours.

Two interventions terminated them reliably. The first was demanding that each proposed route be accompanied by a statement of what would falsify it; a route whose falsification condition could not be stated was almost always a restatement of a previous route. The second was requiring

the system to record what had already been ruled out, and to check new proposals against that record before elaborating them.

The problem of Section 6 illustrates both the pathology and its resolution. Successive reformulations — reshaping the certificate denominator from the irreducible factors of Q ; fixing the telescoper to the operator already known from data and solving only for certificates; splitting the sequence into rational and conjugate branches — were each proposed as breakthroughs and each failed for an independent reason. Only the pole-order computation, which produces a *number* rather than another reformulation, terminated the sequence, by showing that all of them were attempts to outrun a bound of roughly 70,000 unknowns.

3.4 Verification as the system’s own task

A productive pattern, once established, was to direct the system to design checks against its own output rather than to produce more output. Three forms recurred.

Controls with known answers. Before any new machinery was applied to an open problem, it was applied to a solved one. This detected a defective library routine and a sign error in an exponential annihilator, both described in Section 11, either of which would otherwise have been misattributed to the mathematics.

Deliberate negative controls. Where a computation could succeed for the wrong reason, a variant expected to *fail* was run alongside it. In the verification of the inclusion–exclusion representation of Section 5, a control omitting the position layer was included precisely so that agreement of the full assembly could not be attributed to the position layer being inert; the control disagrees exactly when $\mathcal{P} \neq \emptyset$, as required.

Independent re-derivation. Where feasible, a quantity was computed twice by routes sharing no machinery. The Chomp triple was found in one lattice and re-derived in another, then counter-signed by a separately written engine; the cone eigenvalue of Section 8 was obtained both spectrally and from sequence asymptotics; the chain of Section 6 was checked against values computed from the opposite side of the decomposition.

3.5 What the system did not supply

Three things had to come from outside the system.

Cost estimates. The system consistently proposed computations without estimating their runtime, and its estimates when requested were unreliable. Every substantial computation in this project was sized by measurement — a small instance timed, then extrapolated against the known complexity — and several were abandoned on that basis.

Restraint. The system’s disposition is to continue. Judgements that a line of attack was exhausted, that a problem was a trap, or that a result was finished, were the author’s; where they were deferred, effort was consumed without return.

Calibration of evidence. The system reported patterns confirmed on five consecutive values with the same confidence as identities verified symbolically. The distinction drawn in Section 2, and the falsification recorded in Section 4.6, exist because that calibration had to be imposed externally.

3.6 Assessment

The results of Sections 4–10 would not have been obtained by the author alone; the author’s mathematical background at the time of this work comprised a single university course in mul-

tivariable calculus. Equally, they would not have been obtained by the system alone: on the evidence of Section 11, an unsupervised run would have produced a confident and substantially incorrect document. The output is a joint product, and the verification protocol is the mechanism by which the joint product is more reliable than either input.

Part II

Combinatorial game theory

4 The Second Computational Chomp Challenge

4.1 The challenge

Chomp is played on an $a \times b$ grid of cells, rows indexed downward; a move (a *bite*) at cell (i, j) removes every cell (r, c) with $r \geq i$ and $c \geq j$; the player forced to take the poisoned cell $(1, 1)$ loses. Positions are exactly the Young-diagram ideals $\lambda = (\lambda_1 \geq \dots \geq \lambda_a)$.

Gale's strategy-stealing argument [10] shows every rectangular bar other than 1×1 is a first-player win, but certifies no winning move. Computation is the only known access, and Gale already asked whether the winning first move is always unique. The answer is negative — Thompson discovered that the 8×10 bar has two winning moves [4] — but the uniqueness phenomenon is real in low rows: Sheiner [22] has proved that every $3 \times n$ bar has exactly one winning move, settling the three-row case of Gale's question, and Garg [11] has tabulated the $4 \times n$ P -positions to $n \leq 500$.

Ekhad and Zeilberger [7], computing all bars up to 14×14 , found a handful of doubles beyond Thompson's (6×13 , 9×10 , 10×14 , 12×13) and posed as their Second Computational Challenge, with a pledged \$100 donation: find a, b such that the $a \times b$ bar has at least three winning moves.

Answer. The 10×42 bar has exactly three winning bites: at $(5, 36)$, $(7, 30)$, and $(8, 26)$.

4.2 Structural theory of winning bites

Throughout, biting the cell (i, j) of the $a \times b$ bar leaves the two-value Young diagram

$$B(i, j) = (b^{i-1}, (j-1)^{a-i+1}),$$

with rows $1, \dots, i-1$ of length b and rows $i, \dots, a$ of length $j-1$.

Theorem 4.1 (Staircase Theorem). *If (i, j) and (i', j') are both winning bites of the same bar and $i < i'$, then $j > j'$. Moreover no winning bite of an $a \times b$ bar with $a \geq 2$, $ab \geq 2$ lies in row 1; by Proposition 4.2 below, none lies in column 1. Consequently the winning bites form a strict antichain in the product order, there is at most one in each row and each column, and at most $\min(a-1, b-1)$ in total.*

Proof. Suppose $i < i'$ and $j \leq j'$. Bite the cell (i, j) in $B(i', j')$. Rows $1, \dots, i-1$ lie above the bite and retain length b . Rows $i, \dots, i'-1$ have length b in $B(i', j')$, and $b \geq j-1$, so they truncate to $j-1$. Rows $i', \dots, a$ have length $j'-1 \geq j-1$, so they likewise truncate to $j-1$. The result is exactly $B(i, j)$. Two winning bites leave P -positions, and no P -position is one move from another.

For the second claim, a row-1 bite leaves a full $a \times (j - 1)$ rectangle, an N -position by Gale (or, for $j = 2$, a column, which is N for $a \geq 2$). $\square$

The three bites of 10×42 — rows $5 < 7 < 8$, columns $36 > 30 > 26$ — form the first three-step staircase observed. Empirically the staircase is usually tight: in sixteen of the eighteen known doubles, and in the triple, consecutive winning bites are one or two rows apart. It is not always tight; the two side-16 doubles split by six rows (Table 1).

Proposition 4.2 (Transpose closure). *Let $D = (b^p, c^q)$ with $c < b$ and $p + q = a$. Then $D^T = (a^c, p^{b-c})$. The class of two-value diagrams is therefore closed under transposition, the induced map on parameters is the involution $(p, c) \mapsto (c, p)$, and the winning bites of the $a \times b$ bar correspond bijectively to those of the $b \times a$ bar.*

Proof. Column k of D has height equal to the number of rows of length at least k . For $k \leq c$ every row qualifies, giving height $p + q = a$; for $c < k \leq b$ only the p rows of length b qualify, giving height p ; for $k > b$ the height is zero. Applying the map twice returns $(b^p, c^{a-p}) = D$. Transposition is an automorphism of the game fixing the poisoned square, so it preserves P -positions. $\square$

Lemma 4.3 (L -shapes). *The diagram $(b, 1^{a-1})$ is a P -position if and only if $a = b$.*

Proof. The diagram consists of two arms, of lengths $b - 1$ and $a - 1$, meeting at the poisoned square. If the arms are equal, any move shortens exactly one of them and the opponent restores equality; the terminal position has equal arms and is a loss for the player to move. If the arms are unequal, a single move equalises them. $\square$

Theorem 4.4. *The bite at $(2, 2)$ is a winning opening move of the $a \times b$ bar if and only if $a = b$. Consequently the number of winning opening moves of a square bar is **odd**; in particular no square bar has exactly two.*

Proof. Biting $(2, 2)$ leaves $B(2, 2) = (b, 1^{a-1})$, which by Lemma 4.3 is a P -position precisely when $a = b$. For the parity statement take $a = b$. By Proposition 4.2 transposition maps the winning bites of the bar to themselves, acting as $(i, j) \mapsto (j, i)$, with fixed points the diagonal bites. By Theorem 4.1 at most one diagonal bite can be winning, and by the first part $(2, 2)$ is winning. Hence the involution has exactly one fixed point and the remaining winning bites are paired. $\square$

This is consistent with the census below, in which every square through 19×19 has a unique winning move, and it explains the absence of square doubles: a square bar with any off-diagonal winning bite would have at least three.

4.3 Algorithm

One retrograde solve of the ideal lattice of an $A \times B$ box determines the winning-move count of every bar $a \times b$ with $a \leq A$, $b \leq B$, since each such bar is a position in the box. We rank positions by the combinatorial number system on $d_k = \lambda_k + A - k$ (strictly decreasing),

$$\text{rank}(\lambda) = \sum_k \binom{d_k}{A - k + 1},$$

a bijection onto $\{0, \dots, \binom{A+B}{A} - 1\}$ that is monotone under componentwise order, so a single increasing-rank sweep is a valid dynamic-programming order. The children of λ under a bite in

row i at threshold t (all rows $\geq i$ capped at t) have ranks computable in $O(1)$ amortised via the hockey-stick identity

$$\sum_{m=m_1}^{m_2} \binom{t-1+m}{t-1} = \binom{t+m_2}{t} - \binom{t+m_1-1}{t}$$

applied to the capped rows.

The production engine stores one bit per position (P/N), detects N -positions with early exit at the first winning move, recovers exact counts for the bars in a final pass, and parallelises by area layers, all positions of area s depending only on smaller areas. On thirteen cores it solves the $\binom{55}{10} \approx 2.9 \times 10^{10}$ -state 10×45 box — the box in which the triple was found — in under half an hour.

4.4 Validation

Three independent checks were applied.

All engines agree with brute-force game search on six small boxes, both in per-bar winning-move counts and in the full P -position census. The published record is reproduced exactly: Thompson’s 8×10 has two winning moves, at $(4, 9)$ and $(5, 6)$, and the Ekhad–Zeilberger doubles 6×13 and 10×14 check. Finally the triple was found in the 10×45 box, re-derived from the different 10×42 box, and counter-signed by an independent implementation with a different iteration order and no early exit.

4.5 The census

In the searched region — all $a \leq b \leq 19$; $4 \times b$, $b \leq 300$; $5 \times b$, $b \leq 150$; $6, 7 \times b$, $b \leq 110$; $8 \times b$, $b \leq 60$; $9 \times b$, $b \leq 55$; $10 \times b$, $b \leq 45$ — there are exactly eighteen bars with exactly two winning moves and one with three:

$$\begin{aligned} &6 \times 13, 6 \times 93, 7 \times 29, 7 \times 30, 7 \times 57, 8 \times 10, 8 \times 22, 8 \times 23, 9 \times 10, \\ &9 \times 26, 10 \times 14, 10 \times 29, 10 \times 33, 10 \times 35, 11 \times 18, 12 \times 13, 13 \times 16, 14 \times 16 \end{aligned}$$

together with transposes. Every other bar in the region, including every square through 19×19 , has a unique winning move; in particular $4 \times b$ and $5 \times b$ bars are unique throughout their ranges, consistent with — and for four rows implicit in — Garg’s tabulation [11].

Three regularities stand out. *Adjacent pairs*: doubles arrive at consecutive widths ($7 \times 29/30$, $8 \times 22/23$) or consecutive heights (13×16 , 14×16). *Hot and barren lines*: no bar with a side of 15 has a multiple move, while side 16 carries two doubles; row 10 carries six doubles and the triple. *Long gaps*: row 6 is silent from 14 to 92, then hits at 93.

4.6 The structure of multiplicity

The count decomposes. Biting the $a \times b$ bar at (i, j) leaves the two-block position with $p = i - 1$ full rows and $q = a - p$ rows of length $t = j - 1$, so the number of winning moves of the bar equals the number of families (p, q) with $p + q = a$ whose P -track

$$\{(w, t) : (w^p, t^q) \text{ is a } P\text{-position}\}$$

passes through width $w = b$. Empirically each track is sparse and quasi-linear — essentially one P -cell per width — so doubles are crossings of two tracks and the triple is a threefold crossing,

Table 1: Winning bites of every known multi-move bar, in the staircase form guaranteed by Theorem 4.1. The crossing families $(p, q) = (i-1, a-i+1)$ can be read off the bite rows.

bar	bites	bar	bites
6×13	$(4, 12), (5, 9)$	10×14	$(5, 13), (7, 9)$
6×93	$(2, 52), (3, 39)$	10×29	$(6, 16), (8, 12)$
7×29	$(6, 25), (7, 20)$	10×33	$(7, 23), (9, 18)$
7×30	$(4, 21), (5, 20)$	10×35	$(7, 24), (8, 21)$
7×57	$(3, 42), (4, 26)$	11×18	$(9, 17), (10, 16)$
8×10	$(4, 9), (5, 6)$	12×13	$(10, 11), (11, 9)$
8×22	$(2, 9), (3, 7)$	13×16	$(5, 6), (11, 5)$
8×23	$(7, 18), (8, 16)$	14×16	$(3, 12), (9, 10)$
9×10	$(7, 8), (9, 5)$	10×42	$(5, 36), (7, 30), (8, 26)$
9×26	$(2, 11), (4, 10)$		

constrained by Theorem 4.1 to a staircase. The two seven-row doubles, for instance, are the crossings $(5, 2)-(6, 1)$ at $b = 29$ and $(3, 4)-(4, 3)$ at $b = 30$.

Sparsity is provable in the thin-tail case.

Proposition 4.5. *For each (b, p) there is at most one q for which $(b^p, 1^q)$ is a P -position.*

Proof. For $q' < q$ the diagram $(b^p, 1^{q'})$ is reachable from $(b^p, 1^q)$ by removing the cell in row $p + q' + 1$, column 1. The family is totally ordered under the move relation, and no P -position admits a move to a P -position. $\square$

Write $q(b, p)$ for that value of q when it exists. Exact computation gives:

p	$b = 2$	3	4	5	6	7	apparent law
1	1	2	3	4	5	6	$q = b - 1$ (Lemma 4.3)
2	1	2	4	5	7	8	integers not divisible by 3
3	1	3	5	7	9	12	$q = 2b - 3$, false at $b = 7$
4	1	4	7	10	13	14	$q = 3b - 5$, false at $b = 7$

The $p = 1$ row is Lemma 4.3 and is a theorem. The law $q = 2b - 3$ holds for $b = 2, \dots, 6$ — five consecutive values, no exceptions — and fails at $b = 7$, where $q(7, 3) = 12$.

The mechanism is explicit and worth recording. The $p = 3$ slice is almost closed under moves: from $(b^3, 1^q)$ a bite lands either in the same family, or in a rectangle (an N -position by Gale), or in one of exactly two auxiliary families,

$$A(b, c, q) = (b, c, c, 1^q), \quad B(b, c, q) = (b, b, c, 1^q), \quad 1 \leq c < b,$$

arising from bites in rows 2 and 3 respectively. An induction proving $q(b, 3) = 2b - 3$ would therefore require these auxiliary shapes to be N -positions at the relevant parameters. They are not: $B(7, 6, 11)$ is a P -position, so from $(7^3, 1^{11})$ the bite at $(3, 7)$ wins, and $(7^3, 1^{11})$ is an N -position. Direct computation confirms $q(7, 3) = 12$.

The failure is not isolated. The row $p = 4$ exhibits the same pattern: the law $q = 3b - 5$ holds for $b = 2, \dots, 6$ and predicts $q(7, 4) = 16$, whereas direct computation gives 14. Two independent

instances of five consecutive confirmations followed by failure at the sixth value seem to us to settle the status of these apparent laws; the remaining rows are, on present evidence, coincidences. This is consistent with Zeilberger’s conclusion that three-row Chomp is not governed by a periodic structure [26] and with the probabilistic renormalisation analysis of Friedman and Landsberg [8]. We record it as the clearest methodological lesson of the project: five confirmations are not a pattern.

4.7 Quasi-periodicity and rotation numbers

For three-row Chomp the tracks obey laws of unexpected exactness. Writing the n -th P -cell of the family $(2, 1)$ (positions (w, w, t)) as (w_n, t_n) , we find, over 247 cells up to $w = 600$,

$$w_n = (1 + \sqrt{2})n + O(1), \quad t_n = \frac{2+\sqrt{2}}{2}n + O(1),$$

with error confined to a window of width less than 3 and no drift; the family $(1, 2)$ (positions (w, t, t)) hits essentially every t , with $w(t) = \frac{2+\sqrt{2}}{2}t + O(1)$, window less than 2.4 over 351 cells. The slopes’ continued fractions match $\sqrt{2} = [1; \overline{2}]$ and $\frac{2+\sqrt{2}}{2}$ through all computed partial quotients.

Conjecture 4.6. *The diagonal families of three-row Chomp are quasi-periodic with bounded discrepancy and rotation numbers in $\mathbb{Q}(\sqrt{2})$.*

The exact three-row machinery of Brouwer et al. [4], in which Sheiner’s uniqueness theorem [22] is proved, is the natural arena in which to attack this. It is finer than, and consistent with, the renormalisation picture of [8]; Zeilberger’s assessment of three-row Chomp as chaotic [26] referred to the absence of eventual periodicity, which bounded-error quasi-periodicity with irrational rotation number precisely explains.

For the higher families that govern the census the tracks appear to be multi-branch and their arithmetic remains open. The adjacent-pair phenomenon is not a single crossing lingering: the bite rows show that adjacent doubles arise from different family pairs (7×29 crosses $(5, 2)$ – $(6, 1)$ while 7×30 crosses $(3, 4)$ – $(4, 3)$; 8×22 crosses $(1, 7)$ – $(2, 6)$ while 8×23 crosses $(6, 2)$ – $(7, 1)$), so the clustering of crossings at adjacent widths across distinct track pairs is itself the phenomenon demanding explanation. Whether triples occur infinitely often, and where the next one lies, now looks like a question about simultaneous approximation of these rotation numbers.

4.8 The First Computational Chomp Challenge

Ekhad and Zeilberger’s First Computational Challenge, with a pledged \$500 donation, asks for all winning moves of the 1001×1003 bar. We did not solve it, and record here why we believe it to be out of reach by the methods above.

By Theorem 4.1 the problem is to decide, for each of the $\approx 10^6$ candidate bites, whether the resulting two-value diagram is a P -position; equivalently, by the track decomposition, to decide for each of the 1000 families (p, q) with $p + q = 1001$ whether its P -track passes through width 1003. The state space of a direct solve is $\binom{2004}{1001}$, entirely beyond reach.

The structural obstruction is that the two-value diagrams are not closed under all moves: a bite into the protruding rows of (b^p, c^q) produces a three-value diagram, and iterating produces diagrams with unboundedly many distinct row lengths. There is therefore no finite subfamily whose P -positions can be determined internally, and the track laws — the only visible route to extrapolation — are precisely what Section 4.6 shows to be unreliable beyond the computed range.

Part III

Restricted permutations

5 Holonomicity of $a_{r,s}$ and $b_{r,s}$

5.1 Statement

For integers $r, s \geq 1$ set

$$a_{r,s}(n) = \#\{\pi \in \mathfrak{S}_n : \pi_{i+r} - \pi_i \neq s \ (1 \leq i \leq n-r)\}, \quad b_{r,s}(n) = \#\{\pi \in \mathfrak{S}_n : |\pi_{i+r} - \pi_i| \neq s\}.$$

Spahn and Zeilberger [23] ask, as their third challenge with a pledged \$300 donation, whether these sequences are holonomic for all $r, s > 1$, observing that no general theory was available. The case $(1, 1)$ is classical: $a_{1,1}$ satisfies Riordan's recurrence and $b_{1,1}$ is the Hertzprung problem, A002464. Kauers and Koutschan [13] guessed an order-8, degree-11 recurrence for $a_{2,2}$ from 35 terms; proving it is the first challenge of [23], discussed in Section 6.

Theorem 5.1. *For every fixed $r, s \geq 1$, both $a_{r,s}$ and $b_{r,s}$ are holonomic.*

The obstruction to a general result is not the existence of a decomposition but its uniformity: a decomposition whose number of auxiliary variables grows with n yields no holonomicity statement. What we supply is a description of the constraint whose exceptional data stays bounded as $n \rightarrow \infty$.

5.2 The interleaving bijection

Definition 5.2. For $n, k \geq 1$ let $\iota_{n,k}$ list $1, \dots, n$ in the order obtained by concatenating the k arithmetic progressions of step k in increasing order of least element:

$$1, 1+k, 1+2k, \dots, \quad 2, 2+k, \dots, \quad \dots, \quad k, 2k, \dots$$

Interleaving turns a step of k into a step of 1, except where one progression is exhausted and the next begins. We call those places *seams*.

Lemma 5.3 (Seam count). *Let $k \geq 1$ and $n \geq k$. The number of indices $j \in \{1, \dots, n-1\}$ with $\iota_{n,k}(j+1) - \iota_{n,k}(j) \neq k$ is exactly $k-1$. The blocks $P_c = \{c, c+k, c+2k, \dots\} \cap [1, n]$ have lengths*

$$|P_c| = \left\lfloor \frac{n-c}{k} \right\rfloor + 1, \quad c = 1, \dots, k, \quad (1)$$

and the seams occur at the partial sums $|P_1| + \dots + |P_c|$ for $c = 1, \dots, k-1$.

Proof. Within a single P_c consecutive entries differ by exactly k , so no interior index of a block is a seam. A seam can therefore occur only where P_c ends and P_{c+1} begins, and there are exactly $k-1$ such indices. At such an index the difference is $(c+1) - \max P_c$, which is negative and hence not equal to k . Thus every block boundary is a seam. $\square$

Corollary 5.4. *Fix k and c . On each residue class of n modulo k , the seam index $|P_1| + \dots + |P_c|$ is an affine function of n with slope c/k . Consequently, for fixed r, s , all position and value seams are simultaneously affine in n on each residue class of n modulo $\text{lcm}(r, s)$.*

Proof. On a fixed residue class of $n \bmod k$ the floor in (1) equals $(n - c - \rho_c)/k$ for a constant ρ_c , so each $|P_c|$ is affine in n with slope $1/k$, and a partial sum of c of them is affine with slope c/k . The position seams involve $k = r$ and the value seams $k = s$; both are affine on each class modulo $\text{lcm}(r, s)$. $\square$

Now fix $r, s \geq 1$ and n . Define the position relabelling $p = \iota_{n,r}$ and the value relabelling $v = \iota_{n,s}$, and associate to $\pi \in \mathfrak{S}_n$ the permutation $\sigma \in \mathfrak{S}_n$ determined by

$$\sigma_k = v^{-1}(\pi_{p(k)}), \quad k = 1, \dots, n. \quad (2)$$

Since p and v are bijections, (2) is a bijection $\mathfrak{S}_n \rightarrow \mathfrak{S}_n$. Write

$$\mathcal{P} = \{k : p(k+1) - p(k) \neq r\}, \quad \mathcal{V}^+ = \{u : v(u+1) - v(u) \neq s\}, \quad \mathcal{V}^- = \{u : v(u-1) - v(u) \neq -s\}$$

for the position seams and the forward and backward value seams; by Lemma 5.3, $|\mathcal{P}| = r - 1$ and $|\mathcal{V}^+| = |\mathcal{V}^-| = s - 1$.

Proposition 5.5. *Under (2):*

- (i) π satisfies $\pi_{i+r} - \pi_i \neq s$ for all i if and only if σ satisfies $\sigma_{k+1} - \sigma_k \neq 1$ for every k with $k \notin \mathcal{P}$ and $\sigma_k \notin \mathcal{V}^+$;
- (ii) π satisfies $|\pi_{i+r} - \pi_i| \neq s$ for all i if and only if $|\sigma_{k+1} - \sigma_k| \neq 1$ for every $k \notin \mathcal{P}$, where an ascent $\sigma_{k+1} = \sigma_k + 1$ is excused when $\sigma_k \in \mathcal{V}^+$ and a descent $\sigma_{k+1} = \sigma_k - 1$ when $\sigma_k \in \mathcal{V}^-$.

The exceptional data has size $(r - 1) + (s - 1)$ in case (i) and $(r - 1) + 2(s - 1)$ in case (ii); in particular it is bounded independently of n .

Proof. Consider adjacent new indices $k, k+1$ with $k \notin \mathcal{P}$. Then $p(k+1) = p(k) + r$, so the pair corresponds to old positions $i = p(k)$ and $i+r$. Suppose further $\sigma_k = u \notin \mathcal{V}^+$, so $v(u+1) = v(u) + s$. Then $\sigma_{k+1} = u + 1$ if and only if $\pi_{i+r} = v(u+1) = v(u) + s = \pi_i + s$. Outside the excused set the two conditions agree pointwise; on it the adjacency condition is imposed on neither side. This proves (i).

For (ii) the same computation applies to descents with $\mathcal{V}^-$ in place of $\mathcal{V}^+$: with $k \notin \mathcal{P}$ and $u \notin \mathcal{V}^-$ we have $v(u-1) = v(u) - s$, so $\sigma_{k+1} = u - 1$ if and only if $\pi_{i+r} - \pi_i = -s$. Since $|\pi_{i+r} - \pi_i| = s$ means $\pi_{i+r} - \pi_i = \pm s$, both signs must be excused. The cardinalities follow from Lemma 5.3. $\square$

For $(r, s) = (2, 2)$ the excused set has size $1 + 1 = 2$ and Proposition 5.5(i) recovers the bijection of Matsuo as used in [23], where the condition is stated as “ $\pi_{i+1} - \pi_i \neq 1$ except when $i = \lfloor (n+1)/2 \rfloor$ or $\pi_i = \lfloor (n+1)/2 \rfloor$ ”.

5.3 The two-layer inclusion–exclusion representation

Proposition 5.5 reduces both families to a single counting problem. Fix n , a set $\mathcal{P} \subseteq \{1, \dots, n-1\}$ of excused positions and a set $\mathcal{V} \subseteq \{1, \dots, n\}$ of excused values, and let

$$R(n; \mathcal{P}, \mathcal{V}) = \#\{\sigma \in \mathfrak{S}_n : \sigma_{k+1} - \sigma_k \neq 1 \text{ whenever } k \notin \mathcal{P} \text{ and } \sigma_k \notin \mathcal{V}\},$$

with the analogous two-sign version $R^\pm$.

Bonds and blocks

Call a pair (k, v) a *violation* of σ if $\sigma_k = v$, $\sigma_{k+1} = v + 1$, $k \notin \mathcal{P}$ and $v \notin \mathcal{V}$. Inclusion and exclusion over sets of violations gives

$$R(n; \mathcal{P}, \mathcal{V}) = \sum_S (-1)^{|S|} A(S), \quad (3)$$

where S ranges over sets of *value bonds* $S \subseteq \{1, \dots, n-1\} \setminus \mathcal{V}$ and $A(S)$ counts the permutations in which every $v \in S$ occurs as a violation.

Forcing the bonds in S glues the values into maximal runs: if S decomposes $\{1, \dots, n\}$ into $n - |S|$ maximal runs of consecutive integers, a permutation realising all bonds of S is exactly an arrangement of these $n - |S|$ *blocks* in some order. Without the position constraint one would obtain the classical $\sum_S (-1)^{|S|} (n - |S|)!$; the position constraint is what makes the problem two-layered.

The position layer

In an arrangement of blocks the internal bonds occupy the positions determined by the partial sums of the block sizes: if the blocks have sizes $\ell_1, \dots, \ell_{n-|S|}$ in the order used, the internal bonds of the j -th block occupy positions $L_{j-1} + 1, \dots, L_j - 1$ with $L_j = \ell_1 + \dots + \ell_j$. Thus $A(S)$ counts arrangements in which no internal bond position lies in $\mathcal{P}$. Since $|\mathcal{P}| = r - 1$ is bounded, this constraint is removed by a second, finite inclusion–exclusion:

$$A(S) = \sum_{T \subseteq \mathcal{P}} (-1)^{|T|} B(S, T), \quad (4)$$

where $B(S, T)$ counts arrangements in which every position of T is an internal bond position and the positions of $\mathcal{P} \setminus T$ are unconstrained. The sum has $2^{|\mathcal{P}|} = 2^{r-1}$ terms, a number depending only on r .

Block factors

A block of size ℓ absorbs $\ell - 1$ bonds and occupies ℓ consecutive positions, so with x marking positions the unmarked block factor is

$$\mathcal{B}(x) = \sum_{\ell \geq 1} (-1)^{\ell-1} x^\ell = \frac{x}{1+x}. \quad (5)$$

In the two-sign case a block of size $\ell \geq 2$ may be traversed in either direction, and a block of size 1 in only one way, giving

$$\mathcal{B}^\pm(x) = x + 2 \sum_{\ell \geq 2} (-1)^{\ell-1} x^\ell = \frac{x - x^2}{1+x}, \quad (6)$$

which is the block generating function of the Hertzsprung problem — a useful consistency check on the derivation.

Remark 5.6. As a check on (5) and on the umbral normalisation below, take $\mathcal{P} = \mathcal{V} = \emptyset$ and $n = 3$. Then $[x^3] \mathcal{B}^j = 1, -2, 1$ for $j = 1, 2, 3$, and $1 \cdot 1! - 2 \cdot 2! + 1 \cdot 3! = 3$, the number of permutations of $\{1, 2, 3\}$ with no succession, namely 132, 213, 321.

Where the two kinds of mark act

The two excusal types enter in genuinely different ways.

Excused values force block boundaries. If $v \in \mathcal{V}$ then the bond v is never a violation, so $v \notin S$ for every S in (3). Hence no block may span the gap between v and $v + 1$: the excused values cut $\{1, \dots, n\}$ into $|\mathcal{V}| + 1$ *value segments*, and the generating function factorises as a product of one block sequence per segment. A catalytic variable y_i marking the i -th cut records its location.

Excused positions split the arrangement. The position of a bond is determined by the *arrangement* of the blocks, not by their value order, so a position mark cannot be carried inside a block factor. Instead, fixing which arrangement positions $t_1 < \dots < t_q$ are internal bond positions cuts the arrangement into $q + 1$ consecutive *arrangement segments*. Blocks may be distributed among the segments freely and ordered arbitrarily within each, so if the segments receive $j_0, \dots, j_q$ blocks the number of orderings is $j_0! j_1! \dots j_q!$. This is realised by giving each segment its own block-counting variable and applying the multivariate umbral functional

$$\langle \cdot \rangle : z_0^{j_0} \dots z_q^{j_q} \mapsto j_0! \dots j_q!, \quad \langle f \rangle = \int_0^\infty \dots \int_0^\infty f e^{-z_0 - \dots - z_q} dz_0 \dots dz_q. \quad (7)$$

Lemma 5.7. *Let $\mathcal{B}^*(x; u) = \sum_{\ell \geq 2} (-1)^{\ell-1} x^\ell (u + u^2 + \dots + u^{\ell-1})$ be the generating function of a block carrying one distinguished interior slot, u recording the slot's offset from the start of the block. Then*

$$\mathcal{B}^*(x; u) = \frac{-x^2 u}{(1+x)(1+xu)}.$$

Proof. Summing the inner geometric series, $u + \dots + u^{\ell-1} = u(u^{\ell-1} - 1)/(u - 1)$, so

$$\mathcal{B}^* = \frac{u}{u-1} \left(\frac{1}{u} \sum_{\ell \geq 2} (-1)^{\ell-1} (xu)^\ell - \sum_{\ell \geq 2} (-1)^{\ell-1} x^\ell \right).$$

For any y , $\sum_{\ell \geq 1} (-1)^{\ell-1} y^\ell = y/(1+y)$, whence $\sum_{\ell \geq 2} (-1)^{\ell-1} y^\ell = y/(1+y) - y = -y^2/(1+y)$. Substituting $y = xu$ and $y = x$,

$$\mathcal{B}^* = \frac{ux^2}{u-1} \cdot \frac{(1+xu) - u(1+x)}{(1+x)(1+xu)} = \frac{ux^2}{u-1} \cdot \frac{1-u}{(1+x)(1+xu)},$$

since $(1+xu) - u(1+x) = 1-u$; and $(1-u)/(u-1) = -1$. $\square$

Assembling (3), (4), (5) and (7):

Proposition 5.8. *For every n , $\mathcal{P}$ and $\mathcal{V}$,*

$$R(n; \mathcal{P}, \mathcal{V}) = \sum_{T \subseteq \mathcal{P}} (-1)^{|T|} [x^n y_1^{v_1} \dots y_p^{v_p}] \langle \Phi_T(x, z_0, \dots, z_{|T|}, y_1, \dots, y_p) \rangle, \quad (8)$$

where $p = |\mathcal{V}|$, the functional $\langle \cdot \rangle$ is (7), and each Φ_T is a rational function built from (5) — respectively (6) in the two-sign case — as a product of $p+1$ value-segment factors distributed over $|T| + 1$ arrangement segments.

Corollary 5.9. *For fixed r, s the representation (8) is a sum of at most 2^{r-1} terms, each a rational function in at most $1+r$ umbral variables together with at most $s-1$ catalytic variables in the $a_{r,s}$ case and $2(s-1)$ in the $b_{r,s}$ case. All these counts are independent of n .*

5.4 Proof of Theorem 5.1

Fix r, s . By Corollary 5.9, $a_{r,s}(n)$ — respectively $b_{r,s}(n)$ — is obtained from a rational function in a fixed finite set of variables by the following finite sequence of operations, each preserving holonomicity.

(1) *Coefficient extraction.* Extracting the coefficient of $x^n y_1^{v_1} \cdots y_p^{v_p}$ from a rational function in finitely many variables, and more generally taking a diagonal, yields a D-finite series (Lipshitz [16]; Christol). By Corollary 5.9 the number of extractions is at most $1 + p$, independent of n .

(2) *The umbral functional.* Applying $z^j \mapsto j!$ in a variable is integration against e^{-z} , and the resulting parameterised integral of a D-finite integrand is D-finite in the remaining parameters (Almkvist–Zeilberger [2]; Lipshitz [17] for the closure statement). By Corollary 5.9 the functional (7) involves at most $1 + r$ variables.

(3) *Specialisation of the excused data.* By Corollary 5.4 the seam indices are affine in n on each residue class of n modulo $L := \text{lcm}(r, s)$, with slopes c/r and c/s . These slopes are not integers, so we reparametrise: fix $\rho \in \{0, \dots, L-1\}$ and write $n = LM + \rho$. Since $r \mid L$ and $s \mid L$, each seam index becomes an affine function of M with integer slope and integer constant term. Substituting integer-affine functions of M for the catalytic exponents of a holonomic multivariate sequence is a diagonal-type specialisation, and diagonals of D-finite series are D-finite [16]; see also [27, §2] for the holonomic-systems formulation. Hence each of the L sections $M \mapsto a_{r,s}(LM + \rho)$ is holonomic.

(4) *Interlacing.* Finally $a_{r,s}$ is the interlacing of the L sections produced in (3), and an interlacing of finitely many holonomic sequences is holonomic [12, Ch. 7].

Each step preserves holonomicity and the number of steps depends only on (r, s) . $\square$

5.5 Verification

Every component was checked against brute-force enumeration.

The bijection was verified for $(r, s) \in \{(2, 2), (2, 3), (3, 2), (3, 3), (2, 4), (4, 2), (3, 4)\}$ and $4 \leq n \leq 8$, all 32 rows agreeing exactly, with excusal sizes matching Proposition 5.5 in every case. The two-sign case was tested against three candidate excusal rules; only the symmetrised rule of Proposition 5.5(ii) reproduces $b_{r,s}$, the one-sided rule undercounting and a more permissive rule overcounting — the computational signature of both signs requiring excusal. Corollary 5.4 was confirmed for nine (r, s) pairs by fitting each seam index from two points and verifying on the remainder, the fitted slopes agreeing with c/r and c/s .

The representation itself was verified in nine configurations of $(n, \mathcal{P}, \mathcal{V})$ against direct enumeration, computed both as the full assembly (3)–(4) and via the inner inclusion–exclusion over T , with a deliberate control omitting the position layer that disagrees exactly when $\mathcal{P} \neq \emptyset$. Lemma 5.7 was verified symbolically as a truncated bivariate series. Code and data are at [19].

Remark 5.10. Brute force is $O(n!)$, so the checks run to $n = 8$: they confirm a pattern, not a theorem. The theorem is proved above; the computations exist to catch an error in the statement, not to substitute for the proof.

Remark 5.11. This challenge was recorded in our working notes for the entire campaign as research-grade, on the ground that no general theory was known. The computation that settled it required seconds to run. We regard the delay as a failure of the protocol of Section 2, not of the mathematics.

6 The $a_{2,2}$ recurrence: a partial result

The first challenge of [23] asks for a proof of the Kauers–Koutschan order-8, degree-11 operator for $a_{2,2} = \mathbf{A189281}$. We did not obtain it. We record the reduction, which is complete and machine-verified end to end, and the obstruction, which we believe is a statement about our method rather than about the problem.

6.1 The reduction

Matsuo’s bijection (the case $(r, s) = (2, 2)$ of Proposition 5.5) gives

$$a_{2,2}(n) = \text{RIN}(n, \lfloor (n+1)/2 \rfloor, \lfloor (n+1)/2 \rfloor),$$

where $\text{RIN}(n, a, b)$ counts permutations with $\pi_{i+1} - \pi_i \neq 1$ except when $i = a$ or $\pi_i = b$. The two-layer inclusion–exclusion of Section 5 gives $\text{RIN}(n, a, b) = W_a(n) - X(n, a, b)$ with

$$W_a(n) = \langle m! \rangle [G_a(z)G_{n-a}(z)], \quad G_L = [x^L] \frac{1}{1 - zx/(1+x)}.$$

Since $G_L = (-1)^{L-1}z(1-z)^{L-1}$, the W -part has the closed forms

$$e_W(m) = \langle m! \rangle [z^2(z-1)^{2m-2}], \quad o_W(m) = \langle m! \rangle [z^2(z-1)^{2m-3}] \quad (m \geq 2),$$

with values 2, 14, 362, 18806, $\dots$ and 4, 64, 2428, 165016, $\dots$ respectively.

Theorem 6.1. *Both W -parts satisfy order-2 recurrences with polynomial coefficients:*

$$\begin{aligned} -m(2m-1)(2m+3)e_W(m) + \frac{1}{2}(8m^3 + 32m^2 + 32m + 7)e_W(m+1) - \frac{1}{2}(2m+1)e_W(m+2) &= 0, \\ -2(m-1)(m+1)(2m-1)o_W(m) + (4m^3 + 10m^2 + 3m - 1)o_W(m+1) - mo_W(m+2) &= 0. \end{aligned}$$

Both were obtained by creative telescoping in the conjugated form: with $\Omega = (z-1)^2$ and $\lambda = d_z \log \Omega$, one seeks $\sum_i p_i(m)\Omega^i R = (d_z + m\lambda - 1)(N/\Theta)$ with $\Theta = (z-1)^e$ and N constrained to have z -exponent ≥ 1 , which is exactly the boundary condition $N(0) = 0$ needed for the umbral rule $\langle \partial_z P - P \rangle = -P(0)$ to kill the inhomogeneity. Both identities were verified symbolically over $\mathbb{Q}$, together with $N(0, m) = 0$; neither was obtained by sampling. The odd operator appears to be new. Both were subsequently reproduced independently by a patched installation of `ore_algebra`, whose telescoper, after homogenisation by $(g(m)S - g(m+1)) \circ L$, is exactly proportional to the operators above.

6.2 The X -part

The X -part reduces, after elimination of the t -direction by partial fractions, to a diagonal. Writing

$$F_1(v; t) = \frac{(1+v)(1+vt)}{D(v; t)}, \quad F_2(v; t) = \frac{-v^2 t(1+v)(1+vt)}{D(v; t)^2},$$

$$D(v; t) = 1 + (1+t-zt-w)v + t(1-z-w)v^2,$$

the grand generating function of RIN is rational, and $a_{2,2}(2m)$, $a_{2,2}(2m-1)$ are central diagonal coefficients of it. Setting $A_m = [v^m]F_1$ and $B_m = [v^m]F_2$, one has, with $d_1 = 1+t-zt-w$ and $d_2 = t(1-z-w)$:

Proposition 6.2. A_m and B_m are C -finite in m over $\mathbb{Q}(t, z, w)$ with coefficients constant in m :

$$A_k + d_1 A_{k-1} + d_2 A_{k-2} = 0 \quad (k \geq 3),$$

$$B_k + 2d_1 B_{k-1} + (d_1^2 + 2d_2) B_{k-2} + 2d_1 d_2 B_{k-3} + d_2^2 B_{k-4} = 0 \quad (k \geq 5),$$

the second operator being the square of the first, as the denominator D^2 requires.

Since the X -part is $F_2(s; t)F_1(\sigma; t) + F_1(s; t)F_2(\sigma; t)$, the even diagonal collapses to a Hadamard product, and

$$e_X(m) = \langle k! j! \rangle [t^m] (2A_m B_m), \quad o_X(m) = \langle k! j! \rangle [t^m] (B_m A_{m-1} + A_m B_{m-1}). \quad (9)$$

The Hadamard product of C -finite sequences is C -finite with characteristic roots the pairwise products $\{r_1^2, r_1 r_2, r_2^2\}$, each doubled; since $r_1 + r_2 = -d_1$ and $r_1 r_2 = d_2$ are rational, the degree-6 characteristic polynomial $(Y - r_1^2)^2 (Y - r_1 r_2)^2 (Y - r_2^2)^2$ reduces to $\mathbb{Q}(t, z, w)$ by construction rather than by search. Its order, 6, coincides with the order measured independently by fitting an operator to 200 terms of e_X .

Consequently $C_m = A_m B_m$ has a rational bivariate generating function

$$H(x, t; z, w) = \sum_{m \geq 1} C_m(t) x^m = P/Q,$$

with $Q = Q_1^2 Q_2^2$ a perfect square, $\gcd(P, Q) = 1$, and neither factor vanishing at $z = 0$ or $w = 0$. The target becomes the diagonal

$$e_X(m) = \langle k! j! \rangle [x^m t^m] H(x, t; z, w).$$

6.3 Verification of the chain

The chain above reproduces the five independently known exact values $e_X(1, 2, 3) = 0, -4, -48$ and $o_X(2, 3) = -1, -11$, and correctly predicts $e_X(4) = -2100$ and $o_X(4) = -301$, neither used in its construction; these agree with $e_W(4) - a_{2,2}(8) = 18806 - 20906$ and $o_W(4) - a_{2,2}(7) = 2428 - 2729$ computed from the other side of the decomposition. Every intermediate step — the factor recurrences, the degree-6 operator, and $H = P/Q$ — was verified symbolically as an identity of rational functions in (t, z, w) .

6.4 The obstruction

Realising both extractions as residues, with $\Omega = 1/(xt)$, the telescoping identity takes the form

$$\sum_i p_i(m) \Omega^i R = (d_x + m\lambda_x) G_x + (d_t + m\lambda_t) G_t + (d_z - 1) G_z + (d_w - 1) G_w, \quad p_i \in \mathbb{Q}[m].$$

The umbral directions cannot be applied after the diagonal: a recurrence for $[x^m t^m] H$ would have coefficients in $\mathbb{Q}(z, w)$, and $\langle k! j! \rangle$ does not commute with multiplication by z or w . All four directions must therefore be carried simultaneously.

Matching pole orders on the two sides is then decisive. The $i = r$ term contributes a pole at $x = 0$ of order $r + 1$; on the right only the x -direction can raise the x -pole, producing $x^{-(a+1)}$ where $\Theta = x^a t^b Q_1^{c_1} Q_2^{c_2}$. Hence

$$a \geq r, \quad b \geq r, \quad c_1, c_2 \geq 1.$$

At $r = 6$, with the corresponding numerator degrees, this places the ansatz near 70,000 unknowns. Our solver performs dense elimination with cubic cost and a measured rate of $U = 2478$ in 29 minutes; the required size is out of reach by four orders of magnitude.

That this is a statement about the method is shown by what happened next. A one-line defect in `ore_algebra`'s multivariate branch was identified and corrected — the library's own doctests had been failing — and the patched installation reproduced the Hertzprung control operator, then both operators of Theorem 6.1, and then completed the first of the four required eliminations on the full $a_{2,2}$ ideal in seventeen minutes, using Gröbner machinery rather than dense linear algebra. The remaining eliminations were in progress at the time of writing.

Part IV

Solid standard Young tableaux

7 The First Rigorous Challenge

7.1 Statement

A *solid standard Young tableau* of a three-dimensional shape S with N cells is a bijective filling of the cells by $1, \dots, N$ increasing in each of the three coordinate directions. We consider the two-layer shape $S_n = [[n, n], [n, 1]]$, of $3n + 1$ cells, and write $g(n)$ for the number of solid SYT of S_n . The sequence begins

$$2, 48, 1038, 22566, 500144, 11302300, 259808162, 6059911302, \dots$$

Zeilberger computed $g(n)$ for $n \leq 40$, observed empirically that the sequence is P -recursive of order 2 with coefficients of degree 12, and offered a prize for a proof.

Theorem 7.1. Let $K(n) = \frac{4^n(3n)!}{(n+1)!(2n+1)!}$ be the n -th Kreweras number, and set

$$c(n) = \frac{8n^2 + n - 24}{4(n+2)(2n+3)}, \quad T(n) = \frac{7n+5}{5} \prod_{k=1}^n \frac{6(2k-1)(6k-1)(6k+1)}{(k+1)(4k+3)(4k+5)}.$$

Then $g(n) = c(n)(3n+1)K(n) + T(n)$ for all $n \geq 1$. Consequently g lies in the 2-dimensional $\mathbb{Q}(n)$ -module spanned by the hypergeometric terms $(3n+1)K(n)$ and $T(n)$, and therefore satisfies a second-order linear recurrence with polynomial coefficients; the operator, computed by Cramer's rule and displayed in Appendix A, has coefficients of degree 12 and coincides with Zeilberger's empirical recurrence. It has been verified against the first 56 terms.

The route passes through the enumeration of *reverse Kreweras walks*: quarter-plane walks with steps $E = (1, 0)$, $N = (0, 1)$, $D = (-1, -1)$. Writing $k(m; i, j)$ for the number of such walks of length m from the origin to (i, j) , we prove two evaluation theorems of independent interest.

Theorem 7.2 (Diagonal evaluation). For all $n \geq 0$, $\sum_{a=0}^n k(3n-a; a, a) = T(n)$. Moreover

$$k(3n-a; a, a) = K(n) \cdot \frac{(a+1)!(2a+1)!n!(3n-a)!}{4^a(a!)^3(n-a)!(3n)!}, \quad 0 \leq a \leq n,$$

and the diagonal generating function is algebraic:

$$Q_d(x; t) := \sum_{m, a \geq 0} k(m; a, a) x^a t^m = \frac{1}{\sqrt{1 - xT^2}} \left(\frac{T}{2t} + \frac{\sqrt{1 - xT^2} - 1 + \frac{1}{2}xT^2}{tx^2} \right), \quad (10)$$

where $T \equiv T(t)$ is the unique power series with $T = t(2 + T^3)$.

Theorem 7.3 (Ballot-weighted evaluation). *With $\text{Bal}(a, c) = \frac{a-c+1}{a+1} \binom{a+c}{c}$ the ballot numbers,*

$$S(n) := \sum_{0 \leq c \leq a} \text{Bal}(a, c) k(3n - a - c; a, a - c) = k(3n + 1; 1, 0) = \frac{3(9n + 16)(3n + 1)}{8(n + 2)(2n + 3)} K(n).$$

7.2 From tableaux to walks

Order the rows of S_n as $(1, 1), (1, 2), (2, 1)$ (the three rows of length n) and $(2, 2)$ (the single cell). A solid SYT of S_n is equivalent to a growth sequence $\emptyset = P_0 \subset \cdots \subset P_{3n+1} = S_n$ adding one cell at a time. Recording after each step the numbers $x_{11}, x_{12}, x_{21}, x_{22}$ of filled cells in the four rows, the containment constraints are

$$n \geq x_{11} \geq x_{12}, \quad x_{11} \geq x_{21}, \quad x_{22} \leq 1,$$

the cell of row $(2, 2)$ being addable exactly when $x_{12} \geq 1$ and $x_{21} \geq 1$.

Lemma 7.4 (Deletion–insertion). *Solid SYT of S_n are in bijection with pairs (w, τ) where w is a lattice walk of length $3n$ from $(0, 0, 0)$ to (n, n, n) with unit steps in the coordinates (x_{11}, x_{12}, x_{21}) staying in the cone $x_{11} \geq x_{12}, x_{11} \geq x_{21}$, and $\tau \in \{0, \dots, 3n\}$ satisfies $x_{12}(\tau) \geq 1$ and $x_{21}(\tau) \geq 1$.*

Proof. Given a solid SYT, delete the cell of row $(2, 2)$ and record τ as its label minus one, the number of cells added before it. What remains is a growth sequence of the shape $[[n, n], [n]]$, i.e. a walk w as described; the insertion condition at time τ is as stated. The inverse re-inserts the cell after step τ and shifts later labels by one. $\square$

In the difference coordinates $u = x_{11} - x_{12}, v = x_{11} - x_{21}$ the three step types become $(1, 1), (-1, 0), (0, -1)$ and the cone becomes the quarter plane: w is a Kreweras excursion of length $3n$, whence the count $K(n)$ of Kreweras [15]. Complementary counting over the $3n + 1$ slots, using the $x_{12} \leftrightarrow x_{21}$ symmetry, gives:

Lemma 7.5. $g(n) = (3n + 1)K(n) - 2S(n) + T(n)$, where $S(n) = \sum_{\tau} \#\{w : x_{12}(\tau) = 0\}$ and $T(n) = \sum_{\tau} \#\{w : x_{12}(\tau) = x_{21}(\tau) = 0\}$.

Lemma 7.6 (Prefix factorisation).

$$T(n) = \sum_{a=0}^n k(3n - a; a, a), \quad S(n) = \sum_{0 \leq c \leq a} \text{Bal}(a, c) k(3n - a - c; a, a - c).$$

Proof. If $x_{12}(\tau) = x_{21}(\tau) = 0$ the prefix consists of $a := \tau$ steps of type 11 only, leaving the state $(u, v) = (a, a)$; the completion is an arbitrary quarter-plane walk from (a, a) to $(0, 0)$ of length $3n - a$, which on reversing time becomes a reverse-Kreweras walk from the origin to (a, a) . If only $x_{12}(\tau) = 0$, the prefix uses a steps of type 11 and c of type 21 with $x_{11} \geq x_{21}$ throughout — a ballot sequence, $\text{Bal}(a, c)$ choices — ending at $(a, a - c)$. $\square$

7.3 The reverse Kreweras model and two extractions

Let $Q(x, y; t) = \sum_{m,i,j \geq 0} k(m; i, j) x^i y^j t^m$. Constructing walks step by step, a D -step being forbidden on either axis, yields

$$(xy - t(x^2y + xy^2 + 1))Q(x, y) = xy - R(x) - R(y) + c, \quad (11)$$

with $R(x) := tQ(x, 0; t)$ and $c := tQ(0, 0; t)$; here $Q(x, 0) = Q(0, x)$ by symmetry and $+c$ arises from inclusion–exclusion at the origin. The kernel $K(x, y) = xy - t(x^2y + xy^2 + 1)$, as a quadratic in y , has roots with

$$Y_0 Y_1 = \bar{x}, \quad Y_0 + Y_1 = \frac{1}{t} - x, \quad \Delta(x) := (1 - tx)^2 - 4t^2 \bar{x},$$

$Y_0 = \frac{(1-tx) - \sqrt{\Delta(x)}}{2t}$ being the power-series root. The rational kernel $1 - t(x + y + \bar{x}\bar{y})$ is invariant under $(x, y) \mapsto (\bar{x}\bar{y}, y)$ and $(x, y) \mapsto (x, \bar{x}\bar{y})$, generating an orbit of six pairs as in [3, §2.3] and [18, §2.3].

Lemma 7.7 (Validity). *All manipulations of this subsection are identities of formal series in t whose coefficients are Laurent polynomials in x and y . Indeed $[t^m]Q(x, y)$ is a polynomial of degree at most m in each variable, so the substitutions produce elements of $\mathbb{Q}[x, \bar{x}, y, \bar{y}][[t]]$. For the partial-fraction step, factor $K(x, y) = -tx(y - Y_0)(y - Y_1)$; the roots satisfy $Y_0 = t\bar{x} + O(t^2)$ and $1/Y_1 = xY_0$, both in $t\mathbb{Q}[x, \bar{x}][[t]]$, so the expansions $\frac{1}{y-Y_0} = \bar{y} \sum_{k \geq 0} (Y_0 \bar{y})^k$ and $\frac{1}{y-Y_1} = -\frac{1}{Y_1} \sum_{k \geq 0} (y/Y_1)^k$ converge coefficient-wise in t . Hence $[y^0]$ of any product below is well defined and computable term by term; the same applies to the splittings $[\cdot]^{>0}$, $[\cdot]^{\leq 0}$ in x .*

Substituting the pairs (x, y) , $(\bar{x}\bar{y}, y)$, $(x, \bar{x}\bar{y})$ into (11) and forming (first)–(second)+(third), so that only $R(x)$ survives among the unknown univariate series, then dividing by the rational kernel, expanding by partial fractions in y and extracting the constant term in y — only the middle term contributes, producing the diagonal — yields the *diagonal equation*

$$-\bar{x} Q_d(\bar{x}; t) \sqrt{\Delta(x)} = c - \bar{x} - 2R(x) + 2xY_0. \quad (12)$$

Under $x \mapsto \bar{x}$ this is equation (2.4) of [18]; we verified (12) independently to order t^{11} .

Canonical factorisation

The Laurent polynomial $x\Delta(x) = t^2x^3 - 2tx^2 + x - 4t^2$ has one small root $X_0 = 4t^2 + O(t^3)$ and two large ones; writing $\Delta = \Delta_0 \Delta_+(x) \Delta_-(\bar{x})$ in the Kreweras frame one finds, with $T = t(2 + T^3)$,

$$\Delta_0 = \frac{4t^2}{T^2}, \quad \Delta_+(x) = 1 - xT^2, \quad \Delta_-(\bar{x}) = 1 - \bar{x}T\left(1 + \frac{T^3}{4}\right) + \bar{x}^2 \frac{T^2}{4}.$$

Proposition 7.8. $Q_d(x; t)$ is given by (10). In particular $Q_d(0; t) = (4T - T^4)/(8t)$ recovers the excursion generating function.

Proposition 7.9.

$$R(\bar{x}) = \frac{1}{2} \left[c - x + \frac{\bar{x}}{t} - \bar{x}^2 + \left(x + T - \frac{2\bar{x}}{T} \right) \sqrt{\Delta_-(\bar{x})} \right].$$

Proof of Propositions 7.8 and 7.9. Both follow from one splitting. Substituting $2xY_0 = x(1 - tx)/t - (x/t)\sqrt{\Delta(x)}$ into (12) and rearranging,

$$\sqrt{\Delta(x)}\left(\frac{x}{t} - \bar{x}Q_d(\bar{x})\right) = c - \bar{x} + \frac{x}{t} - x^2 - 2R(x), \quad (13)$$

and applying $x \mapsto \bar{x}$,

$$\sqrt{\Delta(\bar{x})}\left(\frac{\bar{x}}{t} - xQ_d(x)\right) = c - x + \frac{\bar{x}}{t} - \bar{x}^2 - 2R(\bar{x}). \quad (14)$$

Dividing (14) by $\sqrt{\Delta_0\Delta_-(\bar{x})}$ yields the *split form*

$$\sqrt{\Delta_+(x)}\left(\frac{\bar{x}}{t} - xQ_d(x)\right) = \frac{c - x + \bar{x}/t - \bar{x}^2 - 2R(\bar{x})}{\sqrt{\Delta_0}\sqrt{\Delta_-(\bar{x})}}. \quad (15)$$

Classify supports in x , per coefficient of t^m . On the left the unknown block $xQ_d(x)\sqrt{\Delta_+(x)}$ involves only powers $x^{\geq 1}$, since $Q_d(x) \in \mathbb{Q}[x][[t]]$ and $\sqrt{1 - xT^2} \in \mathbb{Q}[x][[t]]$. On the right the unknown block $2R(\bar{x})/(\sqrt{\Delta_0}\sqrt{\Delta_-(\bar{x})})$ involves only powers $x^{\leq 0}$.

Positive part. Applying $[\cdot]^{>0}$ kills the R -block:

$$xQ_d(x)\sqrt{\Delta_+(x)} = \left[\sqrt{\Delta_+(x)}\frac{\bar{x}}{t}\right]^{>0} - \left[\frac{T}{2t}\frac{c-x+\bar{x}/t-\bar{x}^2}{\sqrt{\Delta_-(\bar{x})}}\right]^{>0}.$$

Writing $\sqrt{\Delta_+(x)} = \sum_{j \geq 0} \binom{1/2}{j} (-xT^2)^j$, the first extraction is $(\sqrt{1 - xT^2} - 1 + \frac{xT^2}{2})/(tx)$ (the terms $j \geq 2$). In the second, every product of $\{c, \bar{x}/t, \bar{x}^2\}$ with the $\bar{x}$ -series $1/\sqrt{\Delta_-(\bar{x})}$ has only powers $x^{\leq 0}$; the single term reaching a positive power is $(-x) \cdot [\bar{x}^0] \frac{1}{\sqrt{\Delta_-(\bar{x})}} = -x$, contributing $+xT/(2t)$ after the outer sign. Hence

$$xQ_d(x)\sqrt{\Delta_+(x)} = \frac{\sqrt{1 - xT^2} - 1 + \frac{xT^2}{2}}{tx} + \frac{xT}{2t},$$

and dividing by $x\sqrt{\Delta_+(x)}$ gives (10).

Nonpositive part. Applying $[\cdot]^{\leq 0}$ instead kills the Q_d -block:

$$\frac{2R(\bar{x})}{\sqrt{\Delta_0}\sqrt{\Delta_-(\bar{x})}} = \left[\frac{T}{2t}\frac{c-x+\bar{x}/t-\bar{x}^2}{\sqrt{\Delta_-(\bar{x})}}\right]^{\leq 0} - \left[\sqrt{\Delta_+(x)}\frac{\bar{x}}{t}\right]^{\leq 0},$$

where

$$\left[\sqrt{\Delta_+(x)}\frac{\bar{x}}{t}\right]^{\leq 0} = \bar{x}/t - T^2/(2t) \quad (\text{the terms } j \leq 1), \quad \left[-x/\sqrt{\Delta_-(\bar{x})}\right]^{\leq 0} = -x(1/\sqrt{\Delta_-(\bar{x})} - 1).$$

Multiplying by $\frac{1}{2}\sqrt{\Delta_0}\sqrt{\Delta_-(\bar{x})}$ gives Proposition 7.9.

Both propositions were verified against the walk counts to orders t^{15} and t^{13} respectively, and every displayed intermediate identity was checked coefficient-wise. Proposition 7.9 is equivalent to, and corrects a typographical corruption in the published form of, the axis result of [18]. $\square$

7.4 Proof of Theorem 7.2

A reverse-Kreweras walk to (a, a) has length $\equiv 2a \pmod{3}$, so the substitution $x = t$ aligns exponents:

$$A(t) := Q_d(t; t) = \sum_{n \geq 0} \left(\sum_a k(3n - a; a, a) \right) t^{3n} = \sum_{n \geq 0} \tilde{T}(n) t^{3n}.$$

By Proposition 7.8, A is an explicit element of the field $\mathbb{Q}(t, T, \sigma)$ with $\sigma = \sqrt{1 - tT^2}$; eliminating σ and T by resultants gives a minimal polynomial $P(t, A)$ of degree 6 (Appendix B), a polynomial in t^3 , with $P(0, B) = 2B - 2$, so by Hensel's lemma A is the unique power-series root with constant term 1.

Differentiation is explicit in the tower: $T' = (2 + T^3)/(1 - 3tT^2)$ and $\sigma' = -(T^2 + 2tTT')/(2\sigma)$, with reductions $T^3 = (T - 2t)/t$ and $\sigma^2 = 1 - tT^2$. Computing A', A'', A''' in the 6-dimensional basis $\{T^i \sigma^j\}$ and solving a linear system over $\mathbb{Q}(t)$ produces an exact inhomogeneous ODE

$$q_c(t) + q_0(t)A + q_1(t)A' + q_2(t)A'' + q_3(t)A''' = 0$$

with the polynomial coefficients of Appendix B. Every step is exact rational-function arithmetic; as a safeguard the ODE was verified on the series of A to order t^{19} . Extracting coefficients converts the ODE into a recurrence supported on shifts $\{-6, -3, 0\}$ in the exponent of t , hence a second-order recurrence $\sum_{i=0}^2 \gamma_i(n) \tilde{T}(n - i) = 0$ valid for $n \geq 2$. One then checks — an identity of rational functions, verified symbolically — that the hypergeometric ratio

$$\frac{T(m)}{T(m-1)} = \frac{6(2m-1)(6m-1)(6m+1)(7m+5)}{(m+1)(4m+3)(4m+5)(7m-2)}$$

satisfies the same recurrence, that $\gamma_0(n)$ has no integer zeros $n \geq 2$, and that the initial values agree. By induction $\tilde{T}(n) = T(n)$.

For the per- a closed form, extract $[x^a]$ from (10): with $u = xT^2$, $[u^j](1-u)^{-1/2} = \binom{2j}{j} 4^{-j}$ and

$$[u^j] \left(1 - \left(1 - \frac{u}{2} \right) (1-u)^{-1/2} \right) = - \binom{2j-2}{j-1} 4^{-(j-1)} \frac{j-1}{2j} \quad (j \geq 1),$$

an identity of binomial coefficients checked symbolically. Hence

$$[x^a]Q_d = \frac{1}{t} \left[\frac{\binom{2a}{a}}{2 \cdot 4^a} T^{2a+1} - \frac{(a+1)\binom{2a+2}{a+1}}{2(a+2)4^{a+1}} T^{2a+4} \right],$$

and Lagrange inversion, $[t^N]T^k = \frac{k}{N} \binom{N}{(N-k)/3} 2^{N-(N-k)/3}$ at $N = 3n - a + 1$, gives $k(3n - a; a, a)$ as an explicit two-term Gamma expression. That it equals the product form of Theorem 7.2 — equivalently that the ratio in a is $\frac{(a+2)(n-a)(2a+3)}{2(a+1)^2(3n-a)}$ — is a Gamma-quotient identity whose ratio to the claimed form simplifies to 1; verified symbolically, and numerically for all $0 \leq a \leq n \leq 8$. $\square$

7.5 Proof of Theorem 7.3

Let $C(z) = 1 + zC(z)^2$ be the Catalan series. By Lambert's generalised binomial theorem $\sum_{c \geq 0} \text{Bal}(b+c, c) z^c = C(z)^{b+1}$, so the ballot-weighted sum telescopes into a single substitution:

$$\sum_n S(n) t^{3n} = \text{CT}_x \left[C(t^2 \bar{x}) Q(x, t \bar{x} C(t^2 \bar{x}); t) \right].$$

Write $y^* = t\bar{x}C(t^2\bar{x}) = \frac{1-\sqrt{1-4t^2\bar{x}}}{2t}$, a series in strictly negative powers of x satisfying $txy^{*2} - xy^* + t = 0$, i.e. $xy^* = t + txy^{*2}$. Substituting into the kernel gives the collapse

$$K(x, y^*) = xy^* - t(x^2y^* + xy^{*2} + 1) = -tx^2y^*.$$

Since $C(t^2\bar{x})/(ty^*) = x/t^2$, the functional equation (11) yields

$$C(t^2\bar{x})Q(x, y^*) = -\frac{1}{t^2x}(xy^* - R(x) - R(y^*) + c).$$

Taking CT_x : the term xy^* has no x^1 -coefficient, $R(y^*)$ is a series in t and negative powers of x , and c is free of x ; only $R(x)$ survives:

$$\sum_n S(n)t^{3n} = \frac{1}{t^2}[x^1]R(x) = \frac{1}{t} \sum_m k(m; 1, 0)t^m, \quad (16)$$

proving $S(n) = k(3n+1; 1, 0)$. For the closed form, extract $[x^1]$ from Proposition 7.9: with $\beta = 1 + \frac{T^3}{4}$,

$$[x^1]R(x) = \frac{1}{2} \left[\frac{1}{t} - \frac{2}{T} - \frac{T^2\beta}{2} + \frac{T^2}{8}(1 - \beta^2) \right],$$

and under the uniformisation $t = T/(2 + T^3)$ this equals $t^2 \cdot \frac{40t^2 - 18tT + (108t^3 - 1)T^2}{256t^5}$ identically in T (a rational-function identity, verified symbolically). Lagrange inversion then gives

$$S(n) = \frac{1}{256}(-18L(3n+4, 1) + 108L(3n+2, 2) - L(3n+5, 2)) = \frac{3(9n+16)(3n+1)}{8(n+2)(2n+3)}K(n),$$

the last equality a Gamma-quotient identity verified symbolically. $\square$

7.6 The module structure and the recurrence

By Lemmas 7.5, 7.6 and Theorems 7.3, 7.2,

$$g(n) = (3n+1)K(n) - 2S(n) + T(n) = c(n)(3n+1)K(n) + T(n).$$

Both $A(n) := (3n+1)K(n)$ and $T(n)$ are hypergeometric with ratios

$$r_A(n) = \frac{6(3n-1)(3n+1)}{(n+1)(2n+1)}, \quad r_T(n) = \frac{6(2n-1)(6n-1)(6n+1)(7n+5)}{(n+1)(4n+3)(4n+5)(7n-2)},$$

which are not equal as rational functions, so A and T span a free $\mathbb{Q}(n)$ -module of rank 2 closed under the shift. The three vectors expressing $g(n), g(n+1), g(n+2)$ in the basis $(A(n), T(n))$,

$$w_i = \left(c(n+i) \prod_{j=1}^i r_A(n+j), \prod_{j=1}^i r_T(n+j) \right), \quad i = 0, 1, 2,$$

are forced to be linearly dependent; Cramer's rule $\gamma_i = (-1)^i \det(w_j, w_k)$ produces an explicit order-2 operator which, after clearing denominators and content, has degree 12 and is displayed in Appendix A. It agrees up to a scalar with the operator fitted from Zeilberger's data and annihilates all 56 computed terms. $\square$

Remark 7.10. The identity $S(n) = k(3n+1; 1, 0)$ equates a ballot-weighted double sum of general-endpoint walk counts with a single near-axis count. Our proof is a two-line kernel computation; a bijective proof would be desirable, and we pose finding one as an open problem.

Remark 7.11. The closed form for diagonal-endpoint reverse-Kreweras walks complements Kreweras' classical axis formula and the diagonal result of Bousquet-Mélou [3, Thm. 2] for the forward model; we have not found it in the literature. General endpoints provably admit no single hypergeometric closed form of comparable shape.

8 The Second Rigorous Challenge: the cone exponent

The Second Rigorous Challenge asks for a proof that the number $a(n)$ of solid SYT of the cylindrical shape $(2, 1, 1) \times [n]$ is *not* holonomic. Equivalently, $a(n)$ counts walks $(0, 0, 0, 0) \rightarrow (n, n, n, n)$ with positive unit steps in $\mathbb{Z}^4$ confined to the cone $x_1 \geq x_2 \geq x_3, x_1 \geq x_4$. We did not settle it; we record what we established, since it isolates the obstruction sharply.

8.1 Reduction to a spectral question

Since the generating function has integer coefficients and finite radius of convergence, holonomicity would make it a G -function, and by the André–Chudnovsky–Katz theorem its singularity exponents would be rational. The exponent is $\beta = \nu + 3/2$ with

$$\nu = \sqrt{\lambda_1 + \frac{1}{4}} - \frac{1}{2},$$

where λ_1 is the principal Dirichlet eigenvalue of the spherical triangle cut out by the cone, whose walls $x_1 = x_2, x_2 = x_3, x_1 = x_4$ give interior angles $(\pi/3, \pi/2, 2\pi/3)$. Irrationality of ν would establish non-holonomicity.

8.2 What is proved

Domain monotonicity gives a rigorous bracket. The triangle contains a single A_3 Weyl chamber, of angles $(\pi/3, \pi/2, \pi/3)$, for which $\nu = 6$ and $\lambda = 42$; and it is contained in a lune of angle $2\pi/3$, for which $\nu = 3/2$ and $\lambda = 15/4$. Hence

$$\frac{15}{4} < \lambda_1 < 42, \quad \frac{3}{2} < \nu < 6.$$

Its area is $\pi/2$, exactly three A_3 chambers, consistent with the walls generating the full reflection group of S_4 .

The triangle is *not* a reflection fundamental domain, since $2\pi/3$ is not of the form π/k , so the classical rational-exponent formula does not apply. An exhaustive search for polynomial principal eigenfunctions through degree 11 found harmonic multiples of the Vandermonde only at degrees 6, 9, 10, all sign-changing on the interior and hence not principal; the degree-6 one does prove that $\lambda = 42$ lies in the Dirichlet spectrum. No closed form is therefore available.

A Rayleigh–Ritz computation with an admissible trial space gives the rigorous upper bound

$$\lambda_1 \leq 13.7540986066788,$$

and locates $\lambda_2 \approx 26.03$.

8.3 What is computed

Two independent numerical determinations agree:

$$\lambda_1 = 13.7443552132132, \quad \nu = 3.2409029943607.$$

The first is the method of particular solutions in its Betcke–Trefethen subspace-angle form, expanding about the $2\pi/3$ vertex in the Fourier–Legendre basis $P_\nu^{-\mu_k}(\cos \theta) \sin(\mu_k \varphi)$, $\mu_k = k\pi/\alpha$, and collocating the remaining edge; it was calibrated against the A_3 chamber, reproducing $\nu = 6$

to 5×10^{-10} , and successive basis sizes agree to thirteen digits. The second is extrapolation of the sequence asymptotics $a(n) \sim C \cdot 256^n n^{-\beta}$ from a shell-indexed dynamic program.

Two traps in the spectral computation are worth recording, both of which initially produced wrong answers: at integer ν the basis develops identically zero columns, since $P_n^{-m} = 0$ for integer $m > n$, and these must be dropped before the factorisation; and the plain smallest singular value has a spurious floor arising from near-degenerate high-order columns, which completely masked the eigenvalue on the $2\pi/3$ -vertex scan — the Betcke–Trefethen subspace-angle form is essential, not cosmetic.

No rational of denominator at most 60 is compatible with ν ; the nearest, $175/54$, differs by 1.6×10^{-4} , some nine orders of magnitude beyond the numerical uncertainty.

8.4 The obstruction

Excluding rationals of *bounded* denominator suffices only given a bound on the denominator a rational exponent could have. Such a bound would follow from height bounds on a hypothetical annihilating operator: the exponent is a root of the indicial polynomial, whose leading coefficient is controlled by the operator’s coefficient heights, and the rational root theorem then limits the denominator. Kauers and Johansson, in a computation reported in [25, Feb. 28, 2012 update], established from 6000 terms that no recurrence exists with $(\text{ORDER} + 1)(\text{DEGREE} + 1) < 3000$. This bounds the *size* of a hypothetical annihilating operator but not the *heights* of its coefficients, and it is the heights that control the indicial polynomial. We regard closing this gap — effective height bounds for G -operators annihilating a given sequence — as the substantive open problem, and note that the same wall blocks Conjectures 1a and 1b of [14].

Part V

Asymptotics and computation

9 Kauers–Zeilberger Conjectures 2a and 2b

9.1 Statement

Let $G(n)$ be the number of standard Young tableaux of shape $[n, n, n]$ in which every run in every row has length ≥ 2 , and $H(n)$ the analogue for odd run lengths [14]. Conjectures 2a and 2b assert

$$G(n) \sim C_1 \frac{8^n}{n^4}, \quad H(n) \sim C_2 \frac{(7 + 5\sqrt{2})^n}{n^4}.$$

Theorem 9.1. *Both conjectures hold, with $C_1, C_2 > 0$ given by explicit sums over the finitely many admissible boundary configurations. Numerically*

$$C_1 = 0.52128605909(2), \quad C_2 = 0.6389278129(4),$$

both determined in §9.7, the latter computed here for the first time.

Theorem 9.1 is a consequence of a local limit theorem which we believe is of independent interest.

Theorem 9.2 (Excursion LLT for run-modulated tandem walks). *Let R be a run law with $\mathbf{P}(R = r) \propto w_c^r$ on an arithmetic support with exponential tails, tilted to criticality, driving the uniform-on-other-two direction chain with increments Rv_j , where $v_0 = (1, 0)$, $v_1 = (-1, 1)$, $v_2 = (0, -1)$. Let x_0, y_0 be apex-adjacent starting and ending configurations and $\mathcal{L}$ the (position, length) increment lattice. Then, with $V, \widehat{V}$ the discrete harmonic functions of Lemma B and its reverse, $\varrho = [\mathbb{Z}^3 : \mathcal{L}]$ and $\sigma^2 = \text{Var}(R)$,*

$$\mathbf{P}_{x_0, i}(\tau > k, S_k = y_0, L_k = \ell, D_k = d) = \varrho \mathcal{K} V(x_0, i) \widehat{V}(\text{swap } y_0, d) k^{-4} \frac{e^{-(\ell - k\mathbf{E}R)^2/2\sigma^2 k}}{\sqrt{2\pi\sigma^2 k}} (1 + o(1))$$

on the admissible sublattice, and 0 off it.

The proof transports the harmonic-function programme of Denisov–Wachtel [5] and Denisov–Zhang [6] from position-Markov chains to Markov-additive processes. The transport is not formal: the position process fails their pointwise assumptions outright, and the repair — an explicit two-level Poisson corrector ladder — is where the modulated structure is absorbed.

9.2 Lemma A: free-space local CLT

Let $P(\theta, s)$ be the 3×3 Fourier transfer matrix, $P(\theta, s)_{ij} = \frac{1}{2}\varphi(\theta \cdot v_j + s)$ for $j \neq i$, with $\varphi(t) = e^{2it}/(2 - e^{it})$. Verified symbolically: the Perron branch λ satisfies $\lambda(0) = 1$, $\nabla\lambda(0) = (0, 0, 3i)$ — zero position drift, length drift $\mathbf{E}R = 3$ — and

$$-\nabla^2 \log \lambda(0) = \Sigma_3 = \begin{pmatrix} 10/3 & -5/3 & 0 \\ -5/3 & 10/3 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

so the position block is $5M$, the length variance is 2, and the position–length cross-covariance *vanishes*: the length coordinate decouples at Gaussian order. The increment lattice has index 3 in $\mathbb{Z}^3$; on the dual torus $|\varphi(t)| < 1$ for $t \not\equiv 0$ since $|2 - e^{it}| > 1$, so by entry-modulus domination the spectral radius is 1 exactly at the three characters $(\theta, s) = (-s, s, s)$ with $3s \equiv 0 \pmod{2\pi}$; a machine scan gives maximum 0.884 elsewhere.

Uniformly for (z, ℓ) with $(z, \ell - 3k)$ in the admissible coset and $|z| + |\ell - 3k| \leq A\sqrt{k}$,

$$\mathbf{P}_i(S_k = z, L_k = \ell, D_k = d) = \frac{3\pi(d)}{(2\pi k)^{3/2} \sqrt{\det \Sigma_3}} e^{-\frac{1}{2k} Q(z, \ell - 3k)} + o(k^{-3/2}),$$

with the sup-bound $\sup_{z, \ell} \mathbf{P}_i(S_k = z, L_k = \ell, D_k = d) \leq Ck^{-3/2}$.

Proof. Fourier inversion on the torus,

$$\mathbf{P}_i(S_k = z, L_k = \ell, D_k = d) = (2\pi)^{-3} \int_{\mathbb{T}^3} e^{-i(\theta \cdot z + s\ell)} (P(\theta, s)^k)_{id} d\theta ds.$$

Split $\mathbb{T}^3$ into δ -balls around the three unit-modulus characters $\chi_r = (-s_r, s_r, s_r)$, $s_r = 2\pi r/3$, and the complement. On the complement the spectral radius is $\leq 1 - \epsilon(\delta)$ by entry domination and compactness, contributing $O((1 - \epsilon)^k)$. Near $\chi_0 = 0$, $P^k = \lambda^k \Pi + O(\rho_2^k)$ with Π the Perron projector, $\Pi_{id} \rightarrow \pi(d)$ and $\rho_2 < 1$; the expansion $\log \lambda = 3is - \frac{1}{2}Q^*(\theta, s) + O(|\cdot|^3)$, verified symbolically, and the Laplace argument give the Gaussian term with relative error $O(k^{-1/2})$ after $(\theta, s) \mapsto (\theta, s)/\sqrt{k}$. Near χ_r , $r = 1, 2$, the character identity $P(\chi_r + \zeta) = e^{i\psi_r} \text{diag } U_r^{-1} P(\zeta) U_r \text{diag}$, from χ_r being trivial on $\mathcal{L}$, reproduces the same local contribution times $e^{-i\chi_r \cdot (z, \ell)}$, which equals 1 on the admissible coset; summing the three gives the factor $3 = [\mathbb{Z}^3 : \mathcal{L}]$ and cancellation off the coset. The sup-bound follows from the same inversion with absolute values. $\square$

9.3 Lemma B: the corrector ladder and the harmonic function

Denisov and Zhang require the position itself Markov with pointwise martingale structure. Our position fails pointwise — conditional drift $-\frac{3}{2}v_i$, state-dependent covariance — but by *bounded* state functions. The corrector ladder, all verified symbolically, is

$$c(i) = -v_i, \quad q(i) = \frac{8}{3}\left(\frac{W}{3} - v_i v_i^\top\right), \quad r(i) = -\frac{2}{3}v_i,$$

solving $(P - I)c = -\text{drift}$, $(P - I)q = -\widehat{C}$, $(P - I)r = -\widehat{\text{cross}}$. Then $M_k = S_k + c(D_k)$ is an exact martingale, with correction $\leq \sqrt{2}$, and the Poisson-corrected quadratic $|M_k|^2 - k \text{tr} \widehat{C} - \text{tr} q(D_k)$ is a martingale up to a bounded telescoping term. After isotropisation by $(5M)^{-1/2}$ the cone becomes the wedge of angle $\pi/3$, so $p = 3$, and the convex C^∞ geometry assumption holds.

Lemma 9.3 (B.0). $M_k = X_k + c(D_k)$ is a martingale, and so is $N_k := |M_k|^2 - k \text{tr} \widehat{C} - \text{tr} q(D_k)$.

Proof. The first is the c -Poisson equation. For the second, $\mathbf{E}[|M_{k+1}|^2 - |M_k|^2 \mid Z_k = (x, i)] = \text{tr} C(i)$ by martingale increment orthogonality, while the trace of the q -Poisson equation gives $\mathbf{E}[\text{tr} q(D_{k+1}) \mid i] - \text{tr} q(i) = \text{tr} C(i) - \text{tr} \widehat{C}$. Subtract. $\square$

The one place the correctors meet the budget is the f -estimate. Define, on the pair chain,

$$\Lambda(x, i) := u(x) + \nabla u(x) \cdot c(i) + \frac{1}{2} \langle q(i), \text{Hess} u(x) \rangle, \quad f(x, i) := \mathbf{E}_{x,i}[\Lambda(X_1, D_1)] - \Lambda(x, i).$$

Taylor expansion on $\{|\tilde{\Delta}| \leq d(x)/2\}$ kills the order- $|x|^{p-1}$ term by the c -Poisson equation and the order- $|x|^{p-2}$ term by the q -Poisson equation, the stationary part $\frac{1}{2} \text{tr}(\widehat{C} \text{Hess} u)$ vanishing by harmonicity after isotropisation; the remainder is $\leq C|x|^{p-3} \mathbf{E} R^3$ by the third-derivative bound $|\partial^3 u| \leq Cu/d^3$. The tail contributes $\leq C|x|^{p-1} e^{-cd(x)}$ by the geometric majorant. Both sit inside the budget $\beta(x) = |x|^{p-1} d(x)^{-1} \gamma(d(x))$, since $|x|^{p-3}/\beta(x) = d(x)/(|x|^2 \gamma) \leq 1/(d(x) \gamma(d(x))) \downarrow 0$. Hence $|f(x, i)| \leq \tilde{\varepsilon}(d(x)) \beta(x)$.

Lemma 9.4 (B.1, superharmonic envelopes). *There exist $W^\pm(x, i)$, superharmonic respectively submartingale-generating for the killed pair chain on $\{d(x) \geq R\}$, with $W^\pm = u(x)(1 + o(1))$ as $d(x) \rightarrow \infty$ and $W^- \leq \Lambda \leq W^+$.*

Proof. Set $W^\pm = \Lambda \pm G_R(\tilde{\varepsilon}\beta)$ with G_R the Green potential of the cone-killed Brownian motion, for which $G_R(\tilde{\varepsilon}\beta) = o(u)$ by pure cone geometry. Then $\mathbf{E}[W^+(Z_1)] - W^+(x, i) = f(x, i) - \tilde{\varepsilon}\beta(x)$ plus a discrete-versus-continuous potential error; the first two terms are ≤ 0 by the f -estimate, and the discretisation error is absorbed as in [6] using the moment bounds of Lemma C1, an argument using only the Markov property of Z . $\square$

Lemma 9.5 (B.2, the harmonic function). $V(x, i) := \lim_{m \rightarrow \infty} \mathbf{E}_{x,i}[\Lambda(Z_m); \tau > m]$ exists, is harmonic for the killed pair chain, strictly positive on $K^\circ \times \{0, 1, 2\}$, and $V(x, i) = u(x)(1 + o(1))$ as $d(x) \rightarrow \infty$.

Proof. On $\{d \geq R\}$, $W^+(Z_{m \wedge \tau})$ is a nonnegative supermartingale by B.1, so $\mathbf{E}[\Lambda(Z_m); \tau > m]$ is bounded; monotonicity up to the $o(u)$ envelopes gives the limit, and $|V - u| = o(u)$. Uniform integrability follows from $\sup_m \mathbf{E}[|X_{m \wedge \tau}|^p; \tau > m] < \infty$, itself a consequence of Doob applied to W^+ plus the Fuk–Nagaev tail. Harmonicity is the Markov property under the limit. Positivity: for $d(x) \geq R$, $V \geq \Lambda - o(u) \geq u/2$ for R large; for general $(x, i) \in K^\circ$ an explicit route — an r -run in direction 0 followed by an $r/2$ -run in direction 1 moves any point diagonally deep into K while remaining in K — has positive probability and reaches $\{d \geq R\}$. $\square$

Lemma 9.6 (B.3, exit tail and conditional limit). $\mathbf{P}_{x,i}(\tau > m) \sim \varkappa V(x, i) m^{-3/2}$, and conditionally on $\{\tau > m\}$, $X_m/\sqrt{m} \Rightarrow \mu_K$, the u -biased Gaussian on K .

The proof follows the Denisov–Wachtel/Denisov–Zhang scheme. We give it in four steps, and then, in §9.4, itemise precisely which arguments of [6] are imported and what each import requires of the modulated setting.

B.3.0 (functional CLT). $(M_{\lfloor mt \rfloor}/\sqrt{m})_{t \geq 0} \Rightarrow \bar{C}^{1/2} B$. The increments of M are bounded-plus-geometric, square-integrable, and stationary under the uniformly ergodic direction chain, with conditional covariances averaging to $\bar{C}$; the martingale FCLT applies. The correction $|X - M| \leq \sqrt{2}$ is uniformly negligible, so the same holds for X . After isotropisation $\bar{C} = I$.

B.3.1 (Brownian transfer at scale $\sqrt{m}$). For fixed $0 < \delta < A$, uniformly in y with $\delta \leq |y|/\sqrt{m} \leq A$ and $d(y) \geq \delta\sqrt{m}$, and in the state b ,

$$\mathbf{P}_{y,b}(\tau > m) \longrightarrow \mathbf{P}_{y/\sqrt{m}}^{bm}(\tau^{bm} > 1),$$

by B.3.0 and the mapping theorem: the exit functional is Wiener-almost-surely continuous because a Brownian path from an interior point that remains in $\bar{K}$ on $[0, 1]$ but touches ∂K has probability zero, the wedge being convex. The limit is the classical wedge survival probability, with profile $\asymp u(y/\sqrt{m})e^{-c|y|^2/m}$.

B.3.2 (the harmonic-measure limit). For fixed ε ,

$$m^{3/2} \mathbf{E}_{x,i} \left[\mathbf{P}_{X_{\varepsilon m}/\sqrt{m}}^{bm}(\tau^{bm} > 1 - \varepsilon); \tau > \varepsilon m \right] \longrightarrow \varkappa V(x, i) (1 + o_\varepsilon(1)).$$

The Brownian factor equals $\varkappa_1(1 - \varepsilon)^{-3/2} u(z) e^{-|z|^2/2(1 - \varepsilon)}(1 + o(1))$ evaluated at $z = X_{\varepsilon m}/\sqrt{m}$ for z in the bulk, by classical wedge asymptotics, so the left side equals $\varkappa_1 \varepsilon^{-3/2} (\varepsilon m)^{3/2} \mathbf{E}_{x,i}[\tilde{u}(X_{\varepsilon m}); \tau > \varepsilon m]$ for a bounded continuous u -equivalent $\tilde{u}$; by B.2 — uniform integrability of $u(X_{m \wedge \tau})$, the identification $V = \lim \mathbf{E}[\Lambda; \tau > \cdot]$, and $|\Lambda - u| = o(u)$ — this converges to $\varkappa V(x, i)$, with an $o_\varepsilon(1)$ from the Gaussian factor.

B.3.3 (assembly). Apply the Markov property at εm :

$$\mathbf{P}_{x,i}(\tau > m) = \mathbf{E}_{x,i} [\mathbf{P}_{X_{\varepsilon m}, D_{\varepsilon m}}(\tau > (1 - \varepsilon)m); \tau > \varepsilon m].$$

Split the position at time εm . The bulk gives $\text{B.3.1} \times \text{B.3.2} = \varkappa V(x, i) m^{-3/2} (1 + o_\varepsilon(1))$. The near-boundary layer $\{d < \delta\sqrt{m}\}$ contributes $o_\delta(1) \cdot V m^{-3/2}$, by the linear vanishing of the u -profile together with Lemma C2. The far region $\{|y| > A\sqrt{m}\}$ is killed by Lemma C1. Letting $m \rightarrow \infty$, then $\delta \downarrow 0$, $A \uparrow \infty$, $\varepsilon \downarrow 0$, and noting that the constants are consistent across ε by the semigroup property, the limit $\varkappa$ is well defined. The conditional limit follows from the same decomposition applied to $\mathbf{E}_{x,i}[f(X_m/\sqrt{m}); \tau > m]$ for bounded continuous f , using the Brownian meander profile of B.3.1 and dominated convergence via Lemma C2.

Monte Carlo certification over 2M and 4M paths gives tail slope -1.4895 against the predicted -1.5 , wedge width $1.0472 = \pi/3$ to four decimals, and the angular law $\sin 3\varphi$ bin by bin.

9.4 The transported arguments, itemised

The scheme of [5, 6] is written for chains whose *position* is Markov and satisfies pointwise martingale hypotheses. Our position does not: its conditional drift is $-\frac{3}{2}v_i$ and its conditional covariance is state-dependent. Both defects are by *bounded* functions of the modulating state, which is what the corrector ladder exploits. Since the distinction between what is imported and what is re-proved is the substance of the transport, we set it out argument by argument. Throughout, $Z_k = (X_k, D_k)$ is the pair chain, which *is* Markov.

(i) Brownian estimates: imported verbatim. The derivative bounds for the cone-harmonic function u and the Green-function bounds [6, Lemmas 7–9] concern the wedge K and the Brownian motion on it. They involve no chain. After isotropisation by $(5M)^{-1/2}$ our cone is the wedge of angle $\pi/3$ with $p = 3$, convex and C^∞ , so these estimates apply unchanged. What is consumed: only the geometry.

(ii) The f -estimate: re-proved. This is [6, Lemma 10] and the single place where the modulated structure must be absorbed. It is proved above, on the corrected Lyapunov function Λ rather than on u : the order- $|x|^{p-1}$ Taylor term is killed by the c -Poisson equation, the order- $|x|^{p-2}$ term by the q -Poisson equation, and the remainder is bounded by $C|x|^{p-3}\mathbf{E}R^3$ using the third-derivative bound of (i). The conclusion, $|f(x, i)| \leq \tilde{\varepsilon}(d(x))\beta(x)$ with $\beta(x) = |x|^{p-1}d(x)^{-1}\gamma(d(x))$, is verbatim that of [6]; only its proof differs. What is consumed: nothing beyond (i) — this step is self-contained given the ladder.

(iii) Superharmonic construction: imported, with Λ for u . [6, Lemmas 11–14] construct $W^\pm$, establish the existence of the harmonic function as a limit, and prove positivity and the asymptotic $V \sim u$. These arguments operate on u , the auxiliary functions U_β and G , and the f -estimate; they do not otherwise use the chain’s structure beyond the Markov property. Running them on the pair chain with Λ in place of u is therefore legitimate, and since $|\Lambda - u| \leq C|x|^{p-1} = o(u)$ the asymptotics are preserved. V acquires a state argument through c and q . What is consumed: the Markov property of Z , the f -estimate of (ii), and the moment bounds of Lemma C1. *The potential absorption inside the proof of [6, Lemma 12] — the estimate of the discrete-versus-continuous Green potential error — is imported rather than rewritten; it uses only the Markov property and the majorant condition, both available here.*

(iv) Optional-stopping and tail arguments: imported. [6, §4–6] derive the exit tail and the conditional limit. Their optional-stopping steps require exact martingales; ours are supplied by B.0, with bounded telescoping terms absorbed into the constants. Their functional CLT input is replaced by the martingale FCLT for finitely-modulated Markov-additive processes (B.3.0). Their Fuk–Nagaev input is Lemma C1. What is consumed: B.0, B.3.0, C1, and the truncation argument of [6, Prop. 19], the last imported directly.

Status. With (i)–(iv) and B.0–B.3 the transported form of [6, Theorem 6] is complete, and with it Lemma B. We record explicitly that three components — the potential absorption within (iii), portions of the replay within B.3.3, and the truncation import within (iv) — are citations rather than rewritten pages. Every model-specific hypothesis those arguments require is verified in [20]: the Perron drift and covariance structure, the torus aperiodicity scan, the corrector equations, and the majorant condition.

9.5 Uniform local bounds and the excursion theorem

Lemma C1 is Fuk’s martingale inequality applied coordinatewise to $(M_k, L_k - 3k)$, the corrected chain having increments $\tilde{\Delta} = (R - 1)v_j + v_i$ with $|\tilde{\Delta}| \leq \sqrt{2}R$, so the majorant $Y = \sqrt{2}R$ is geometric with all moments finite. Lemma C2 is the uniform local upper bound, obtained by splitting at $\lfloor m/2 \rfloor$ and combining the exit-tail bound with the sup-form of Lemma A.

Lemma D assembles these. The reversed increments $\{-v_i\}$ form the transpose tandem, so $\widehat{\text{walk}} = \text{walk} \circ \text{swap}$ and $\widehat{V} = V \circ \text{swap}$; the direction chain is reversible with uniform stationary

law, so all reversal Radon–Nikodym factors are 1. The exact split-and-reverse identity

$$\begin{aligned} & \mathbf{P}_{x_0,i}(\tau > 2m, S_{2m} = y_0, D_{2m} = d) \\ &= \sum_{w,c} \mathbf{P}_{x_0,i}(\tau > m, S_m = w, D_m = c) \hat{\mathbf{P}}_{y_0,d}(\hat{\tau} > m, \hat{S}_m = w, \hat{D}_m = c) \end{aligned}$$

combines with the bulk-local theorem — proved by splitting at $m - \lfloor \varepsilon m \rfloor$, using Lemma A on the final window, bounding near-boundary starts by $O(\eta/\varepsilon)$ with $\eta = \varepsilon^{2/3}$, and bounding re-entry by $O(e^{-c\varepsilon^{-1/3}})$ — to give Theorem 9.2.

9.6 The exact tilt identity and the constants

By the lattice-word bijection, $G(n)$ counts ballot words on $\{0, 1, 2\}$ with n letters each in which every maximal run has length ≥ 2 ; reading the word as tandem steps turns these into quarter-plane excursions of length $3n$ with the same run restriction. Within a run the position moves monotonically along a fixed v_d , and the decreasing coordinate attains its minimum at the run's end, so a path lies in K at every unit step if and only if it lies in K at every run boundary: the unit-step cone constraint is exactly the MAP cone constraint.

Proposition 9.7 (Criticality as k -independence). *Fix a run law $\mathbf{P}(R = r) = w^r/W(w)$ on support S with $W(w) = \sum_{r \in S} w^r$. An excursion with k runs of lengths $r_1, \dots, r_k$ has MAP-probability*

$$\left(\frac{1}{2}\right)^{k-1} \prod_i \frac{w^{r_i}}{W(w)} = 2 w^{3n} \left(\frac{1}{2W(w)}\right)^k.$$

At the critical point $2W(w_c) = 1$ the bracket is 1 and the weight is $2w_c^{3n} = 2\mu^{-n}$, independent of k and of the run profile.

For the ≥ 2 law, $W = w^2/(1-w)$, $w_c = \frac{1}{2}$, $\mu = 8$; for the odd law, $W = w/(1-w^2)$, $w_c = \sqrt{2} - 1$, $\mu = (1 + \sqrt{2})^3 = 7 + 5\sqrt{2}$. Hence exactly

$$G(n) = \frac{8^n}{2} \sum_k \mathbf{P}_0(\tau > k, S_k = 0, L_k = 3n),$$

and identically for $H(n)$.

Applying Theorem 9.2 to each term, with apex configurations regularised by one-step harmonicity $V(\mathbf{0}) := \sum_r \mathbf{P}(R = r)V((r, 0), 0) < \infty$, and using the reversal-swap identity to give $\hat{V}(\text{swap } \mathbf{0}) = V(\mathbf{0})$, the length Gaussian sums over k to contribute $1/\rho$ with $\rho = \mathbf{E}R$. Altogether

$$C = \frac{\varrho \mathcal{K}}{2\rho} V(\mathbf{0})^2 : \tag{17}$$

the excursion constant is, up to a universal factor, the *square* of the harmonic function's apex value.

For any run law with exponential tails under the critical tilt, the lag-correlation collapse gives $\Sigma = (\mathbf{E}R^2 - \frac{2}{3}(\mathbf{E}R)^2)M$, a scalar multiple of M , so the isotropised cone is always the $\pi/3$ wedge, $p = 3$, and the excursion exponent is -4 . For the odd-run model this is machine-certified at $w_c = \sqrt{2} - 1$: $\mathbf{E}R = \sqrt{2}$, $\mathbf{E}R^2 = 3$, $\Sigma_H = \frac{5}{3}M$ with the Perron computation cross-checked, corrector $c_H(i) = -\frac{\sqrt{2}}{3}v_i$, growth $\mu_H = 7 + 5\sqrt{2}$. Theorem 9.1 follows for both conjectures

simultaneously. Numerically, from independent dynamic programs validated against brute force and calibrated against the known C_1 with the same estimator and n -range, error $\lesssim 4 \times 10^{-7}$:

$$C_1 = 0.52128605909(2), \quad C_2 = 0.6389278129(4),$$

both as established in §9.7 below. No closed form is known for either; PSLQ searches over natural constant bases have found nothing.

9.7 Numerical determination of the constants

Kauers and Zeilberger [14] estimate C_1 “close to 0.521286” and C_2 “close to 0.63892”, to five and six significant digits respectively. Since (17) identifies each constant with the square of a harmonic-function value for which no closed form is known, the constants must be obtained numerically, and we determined both to higher precision than the literature records.

Enumeration. Both sequences were computed exactly by a layered dynamic program on the lattice-word model of §9. States are (c_1, c_2, d, r) at total word length s , where $c_0 = s - c_1 - c_2$, the letter $d \in \{0, 1, 2\}$ is the last written, and r records the current run length capped so as to retain both its parity and whether it exceeds one; the ballot condition $c_0 \geq c_1 \geq c_2$ is maintained as an invariant, and a change of letter is admissible only when the current run length is admissible for the model — at least two for G , odd for H . All arithmetic is exact.

The program reproduces the published initial terms of both sequences. For G , all seventeen terms

$$0, 1, 1, 5, 15, 69, 304, 1518, 7807, 42314, 236621, \\ 1364570, 8062975, 48680547, 299388670, 1871463427$$

(from $n = 1$); for H , the twelve published terms

$$1, 2, 9, 46, 306, 2252, 18308, 158872, 1454570, 13888112, 137277741, 1396638636.$$

It was run to $n = 500$ for G , where $G(500)$ has 441 decimal digits, and to $n = 200$ for H .

Acceleration. Writing $A_N = a(N) N^4 \mu^{-N}$ with $\mu = 8$ respectively $\mu = 7 + 5\sqrt{2}$, so that A_N tends to the constant, and assuming the expansion $A_N = C(1 + a_1/N + a_2/N^2 + \cdots)$, iterated Richardson extrapolation

$$A_N^{(k+1)} = \frac{(N+1)A_{N+1}^{(k)} - (N-k)A_N^{(k)}}{k+1}$$

was applied. As an independent check under different hypotheses, Salzer’s transformation was applied to the same data; and a third estimate was obtained by fitting the increments of the deepest Richardson column, which decay geometrically, and summing the tail.

Results. For C_1 , at $n_{\max} = 500$ and Richardson order 28:

$$\begin{array}{ll} \text{Richardson:} & 0.5212860590903885965 \\ \text{Salzer:} & 0.5212860590836008916 \\ \text{tail fit:} & 0.5212860590914060289 \end{array}$$

agreeing to 7.8×10^{-12} , with fitted decay ratio 0.846. For C_2 , at $n_{\max} = 200$ and Richardson order 14:

Richardson:	0.6389278129267359401
Salzer:	0.6389278129316865333
tail fit:	0.6389278129455990734

agreeing to 1.9×10^{-11} , with fitted decay ratio 0.650. We therefore record

$$C_1 = 0.52128605909(2), \quad C_2 = 0.6389278129(4), \quad (18)$$

in each case the digits shown being those on which the three accelerations agree, with the final parenthesised digit bracketed to within 10^{-11} . Both refine the published estimates, which are correct as far as they are stated. Note that C_2 is obtained here *directly*, from its own enumeration, and not by calibration against C_1 ; the two determinations are independent.

A cautionary note on extrapolation. An intermediate determination of our own, from the same enumeration for G but carried only to $n = 110$ with Richardson order six, gave 0.5212860373, which is in error from the eighth significant digit. The cause is visible in the convergence table: the Richardson column ascends monotonically and at order six has not resolved that digit, continuing to rise for a further twenty orders. Under-extrapolation of a monotone sequence produces a value that appears converged and is not, and the error was detected only when the figure was recomputed from scratch. We record this because the same failure is available to anyone using acceleration without an independent second method, and because the agreement of Richardson with Salzer — which converges under different hypotheses — is what makes (18) trustworthy where a single method would not be.

10 The Geode Challenge

10.1 Statement

Let $S = 1 + \sum_{k \geq 2} t_k S^k$ be the hyper-Catalan generating series of Wildberger–Rubine, restricted to D variables $t_2, \dots, t_{D+1}$, and let G be the Geode, $S - 1 = (t_2 + \dots + t_{D+1})G$. Write $G[M^{(D)}]$ for the coefficient of $(t_2 \cdots t_{D+1})^M$ in G . Amdeberhan, Kauers and Zeilberger [1] offered a donation for $G[1000^{(5)}]$; Rubine [21], who computed the four-dimensional value in 35.9 hours, estimated the five-dimensional case at roughly a thousand times that cost and wrote that new innovations were required.

The hyper-Catalan coefficients have the Erdélyi–Etherington closed form

$$C[\mathbf{m}] = \frac{K!}{(1 + K - N)! \prod_k m_k!}, \quad K = \sum_k k m_k, \quad N = \sum_k m_k.$$

Expanding $1/(t_2 + \dots + t_{D+1})$ geometrically in t_2 gives, with $d = D - 1$,

$$G[M^{(D)}] = \sum_{\alpha \in \{0, \dots, M\}^d} (-1)^{|\alpha|} \binom{|\alpha|}{\alpha} C[M+1+|\alpha|, M-\alpha_1, \dots, M-\alpha_d], \quad (19)$$

a sum with $(M+1)^d$ terms — 10^{12} at $M = 1000$, $D = 5$: the wall described by Rubine.

10.2 The collapse

Write $W = \sum_{j=1}^d j\alpha_j$, $b_K = M \sum_{k=2}^{D+1} k + 2$ and $b_N = M \sum_{k=1}^D k + 1$.

Lemma 10.1. *For every α in (19), the vector $\mathbf{m}'$ appearing in $C[\cdot]$ satisfies $N' = DM + 1$, independent of α , and $K' = b_K - W$.*

Proof. $N' = (M+1+|\alpha|) + \sum_j (M - \alpha_j) = DM + 1$. For K' , the shift of m_2 contributes $2|\alpha|$ while the decrements remove $\sum_j (j+2)\alpha_j = W + 2|\alpha|$. $\square$

Consequently $K'/(1 + K' - N)! = (K')_{DM} = (DM)!(\frac{K'}{DM})$, a polynomial in W , and $(\frac{K'}{DM}) = [z^{DM}](1+z)^{b_K-W}$.

Lemma 10.2. *With $T[i, W] = \sum_{|\alpha|=i, W(\alpha)=W} \prod_j \binom{M}{\alpha_j}$, $\sum_{i,W} T[i, W] u^i y^W = \prod_{j=1}^d (1 + uy^j)^M$.*

Lemma 10.3. $\frac{i!}{(M+1+i)!} = \frac{1}{M!} \int_0^1 x^i (1-x)^M dx$.

Substituting Lemma 10.1 into (19), using $1/(\alpha_j!(M - \alpha_j)!) = \binom{M}{\alpha_j}/M!$, then evaluating the α -sum by Lemma 10.2 at $u = -x$ via Lemma 10.3:

Theorem 10.4. *With $d = D - 1$, $s_d = d(d+1)/2$, $E = (\sum_{k=2}^{D+1} k - s_d)M + 2$ and $h(x, z) = (1-x) \prod_{k=1}^d ((1+z)^k - x)$,*

$$G[M^{(D)}] = \frac{(DM)!}{(M!)^D} [z^{DM}] (1+z)^E \int_0^1 h(x, z)^M dx.$$

10.3 Algorithm and results

Modulo a prime p , the integral of the polynomial $h(\cdot, z)^M$, of x -degree DM , is evaluated exactly by interpolatory quadrature at $DM + 1$ nodes avoiding $x = 1$; at equispaced nodes the Lagrange denominators are $(-1)^{n-1-j} j!(n-1-j)!$. For each node $g_j(z) = h(x_j, z)$ has degree s_d , and the coefficients of g_j^M satisfy the order- s_d recurrence obtained from $gP' = Mg'P$, costing $O(s_d)$ operations per coefficient; they are contracted on the fly against $\binom{E}{DM-r}$ and the quadrature weights.

The cost is $O(D^2 s_d M^2)$ arithmetic operations and $O(DM)$ memory per prime, against $O(M^3)$ and $O(M^2)$ for the naive evaluation; roughly 10^3 to 1.6×10^3 thirty-bit primes reconstruct the values by the Chinese remainder theorem, at a measured 2.64 seconds per prime for $D = 5$, $M = 1000$. A proven bound

$$|G[M^{(D)}]| \leq \frac{b_K!}{(1+b_N)!(M!)^d} \sum_i \frac{i! \binom{dM}{i}}{(M+1+i)!},$$

from $\sum_W T[i, W] = \binom{dM}{i}$ and the monotonicity of $W \mapsto (b_K - W)_{DM}$, sizes the reconstruction.

We report $G[1000^{(D)}]$ for $D = 5, \dots, 10$, all computed for the first time; the five-dimensional value has 8367 digits. Validation is threefold: the $D = 4$ run reproduces Rubine's published value exactly, in under three minutes; the small values $H(1), \dots, H(6)$ agree with the printed values of [1], confirming object identity; and every reconstructed value is checked against fresh primes not used in the reconstruction. We note that the prose of [21] reports a digit count inconsistent with

the growth rate of the smaller values tabulated there; the digits themselves are consistent with our computation.

The four-variable recurrences posed as an open problem in [1] remain open. We observe that the bottleneck the authors report — term generation — is removed by Theorem 10.4, so the problem may now be within reach of a guessing computation.

Part VI

Assessment

11 Negative results and errors

We regard the following record as an integral part of the report. Every error below was detected by comparison with an independently known quantity; none was detected by inspection.

11.1 Errors in generated code

A degree cap computed from an incorrect bound silently corrupted a sequence beyond $n \approx 104$. The corruption was invisible at small n and was caught only by fitting an operator to the full sequence — the lesson being that the operator-fit control on the complete sequence, not agreement at small n , is the load-bearing gate for extended data.

Floating-point division inside a compiled rational function destroyed every matrix entry. The resulting noise matrix had full rank, which presented as “no solution at any ansatz size” — a false negative indistinguishable from a genuine one without the instrumentation of Section 2.

Integer coercion silently truncated rational coefficients, causing a correct operator with half-integer coefficients to be reported as failing its own data check.

11.2 Errors of sizing

Several computations were attempted without estimating their cost. A symbolic construction requiring only six coefficients was written to compute twelve and did not terminate. A solver evaluated a large rational function once per matrix column rather than once per sample point, incurring a factor equal to the number of columns; restructuring reduced a run from forty-seven seconds to under one at the same problem size, and made a subsequent stage feasible.

11.3 Errors of formulation

A sign error in the annihilator of an exponential factor produced two days of subtly wrong output. It was diagnosed only when the discrepancy was recognised as exactly twice a boundary term — the signature of a mis-specified exponential — rather than as a random disagreement.

A decomposition of a C -finite sequence into rational and conjugate branches was verified as mathematics and then found inapplicable: the branches are rational rather than polynomial in the umbral variables, while the umbral functional is defined only on polynomials. The mathematics was correct and the application was not, a failure mode we found particularly hard to detect.

11.4 Errors of conjecture

The thin-tail track laws of Section 4.6 each held on five consecutive values and are false. We know of no better illustration of the distinction between *supported* and *machine-verified* drawn in Section 2.

12 Concluding remarks

Five challenge problems were solved and two reduced. Beyond the solutions the project yielded results we believe to be of independent interest: the Staircase Theorem and the parity theorem for square Chomp bars; the explicit algebraic generating function for reverse-Kreweras diagonal walks together with the closed form for diagonal-endpoint counts; the identity $S(n) = k(3n + 1; 1, 0)$; the order-2 recurrence for the odd W -part of $a_{2,2}$; the integral representation for diagonal Geode coefficients; the local limit theorem for excursions of Markov-modulated walks in cones and the square identity (17) for its constants; and two-sided information on the cone exponent of the $(2, 1, 1) \times [n]$ shape.

The methodological conclusion is that the mode of work described in Section 1.2 is productive but requires a verification discipline more stringent than is customary, because the characteristic failure is not absence of output but plausible incorrect output. The protocol of Section 2 is, in our experience, the minimum sufficient response; the record of Section 11 is what it caught.

Acknowledgements

We thank Doron Zeilberger for posing the problems, for his correspondence throughout, and for suggesting that this work be collected into a single paper; Manuel Kauers and Doron Zeilberger for arXiv endorsements; and Shalosh B. Ekhad, whose tables calibrated the Chomp work. The analytic backbone of Section 9 follows the work of Denisov, Wachtel and Zhang, without which the transport performed there would have had nothing to ride on. The AI system used throughout was Claude, developed by Anthropic.

A The order-2 operator for $g(n)$

The recurrence is $p_0(m)g(m) + p_1(m)g(m-1) + p_2(m)g(m-2) = 0$ for $m \geq 3$, with:

$$\begin{aligned} p_0(m) = & 258048m^{12} + 1169664m^{11} + 326816m^{10} - 5324160m^9 \\ & - 6298026m^8 + 6420627m^7 + 12567093m^6 - 643740m^5 \\ & - 8475047m^4 - 2174541m^3 + 1536066m^2 + 353700m \\ & - 113400 \end{aligned}$$

$$\begin{aligned} p_1(m) = & -13934592m^{12} - 28325376m^{11} + 80053056m^{10} + 126497520m^9 \\ & - 162057066m^8 - 177863088m^7 + 133677768m^6 + 95456580m^5 \\ & - 43376892m^4 - 16667436m^3 + 5184126m^2 + 334800m \\ & - 113400 \end{aligned}$$

$$\begin{aligned}
p_2(m) = & 188116992m^{12} - 87899904m^{11} - 1694242656m^{10} + 1930644720m^9 \\
& + 3397410216m^8 - 5838018732m^7 - 342563508m^6 + 4226543280m^5 \\
& - 1437148908m^4 - 737373564m^3 + 448283664m^2 - 62823600m \\
& + 6350400
\end{aligned}$$

B The exact ODE for the diagonal series

With $A(t) = Q_d(t; t)$: $q_c + q_0A + q_1A' + q_2A'' + q_3A''' = 0$, where:

$$q_c(t) = -81 (114t^3 - 5)$$

$$q_0(t) = 81 (3990t^6 + 282t^3 - 5)$$

$$q_1(t) = t (646380t^6 + 7845t^3 - 583)$$

$$q_2(t) = 24t^2 (10773t^6 - 99t^3 - 8)$$

$$q_3(t) = 16t^3 (3t - 1) (57t^3 + 1) (9t^2 + 3t + 1)$$

C Numerical data

All values below are exact integers unless stated otherwise, and each was computed by at least two independent routes.

D.1 Solid SYT of shape $[[n, n], [n, 1]]$

n	$g(n)$
1	2
2	48
3	1038
4	22566
5	500144
6	11302300
7	259808162
8	6059911302

The full list to $n = 55$, computed by two independent dynamic programs which agree throughout, is in the repository of Appendix D; the operator of Appendix A annihilates all 56 terms.

D.2 Restricted permutation counts

The counts $a_{r,s}(n)$ and $b_{r,s}(n)$ for $4 \leq n \leq 8$, computed by direct enumeration of $\mathfrak{S}_n$ and reproduced exactly by the transformed counts of Proposition 5.5:

(r, s)	$n = 4$	5	6	7	8
$a_{r,s}(n)$					
(2, 2)	18	75	410	2729	20906
(2, 3)	20	88	480	3082	23232
(3, 2)	20	88	480	3082	23232
(3, 3)	22	98	534	3414	25498
(2, 4)	— — —	102	544	3480	25944
(4, 2)	— — —	102	544	3480	25944
(3, 4)	— — —	108	588	3768	28032
(4, 3)	— — —	108	588	3768	28032
(4, 4)	— — —	114	628	4062	30360
$b_{r,s}(n)$					
(2, 2)	16	44	200	1288	9512
(2, 3)	16	64	336	1776	12224
(3, 2)	16	64	336	1776	12224
(3, 3)	20	80	384	2240	15424
(2, 4)	— — —	84	400	2352	16512
(4, 2)	— — —	84	400	2352	16512
(3, 4)	— — —	96	480	2784	19200
(4, 3)	— — —	96	480	2784	19200
(4, 4)	— — —	108	544	3264	23040

Here $a_{2,2} = \text{A189281}$ and $b_{2,2} = \text{A110128}$. The equalities $a_{r,s} = a_{s,r}$ and $b_{r,s} = b_{s,r}$ visible in the table are instances of the transpose symmetry.

D.3 Excusal sizes

The sizes of the exceptional data of Proposition 5.5, confirmed constant in n over $4 \leq n \leq 8$ in every case:

(r, s)	(2, 2)	(2, 3)	(2, 4)	(3, 2)	(3, 3)	(3, 4)	(4, 2)	(4, 4)
$(r - 1) + (s - 1)$	2	3	4	3	4	5	4	6
$(r - 1) + 2(s - 1)$	3	5	7	4	6	8	5	9

D.4 The decomposition of $a_{2,2}$

The W - and X -parts of Section 6, by parity:

m	$e_W(m)$	$e_X(m)$	$o_W(m)$	$o_X(m)$
1	2	0	—	—
2	14	−4	4	−1
3	362	−48	64	−11
4	18806	−2100	2428	−301
5	—	−146704	165016	—
6	—	−15903636	—	—

The identities $e_W(m) - e_X(m) = a_{2,2}(2m)$ and $o_W(m) - o_X(m) = a_{2,2}(2m - 1)$ provide the cross-check used in Section 6: at $m = 4$, $18806 - (-2100)$ is not the relevant combination; rather

$e_X(4) = e_W(4) - a_{2,2}(8) = 18806 - 20906 = -2100$ and $o_X(4) = o_W(4) - a_{2,2}(7) = 2428 - 2729 = -301$, both of which the chain of (9) predicted before being shown them.

D.5 Thin-tail Chomp tracks

The unique q with $(b^p, 1^q)$ a P -position, in the notation of Proposition 4.5:

$p \backslash b$	2	3	4	5	6	7
1	1	2	3	4	5	6
2	1	2	4	5	7	8
3	1	3	5	7	9	12
4	1	4	7	10	13	14
5	1	5	7	13	15	19
6	1	5	10	9	16	—

The boldface entries are the two falsifications discussed in Section 4.6: the law $q = 2b - 3$ predicts $q(7, 3) = 11$ and the law $q = 3b - 5$ predicts $q(7, 4) = 16$, against the computed 12 and 14.

D.6 Two-value Chomp grids

The P -cells (p, c) of the two-value diagrams (b^p, c^q) with $p + q = a$, for small bars. Each grid contains exactly one cell, in accordance with the uniqueness of the winning move for these sizes, and the transpose symmetry $(p, c) \mapsto (c, p)$ is visible on comparing $a \times b$ with $b \times a$:

$a \backslash b$	2	3	4	5	6	7
2	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
3	(2, 1)	(1, 1)	(1, 2)	(2, 3)	(1, 3)	(2, 4)
4	(3, 1)	(2, 1)	(1, 1)	(2, 2)	(1, 2)	(2, 3)
5	(4, 1)	(3, 2)	(2, 2)	(1, 1)	(2, 3)	(1, 2)
6	(5, 1)	(3, 1)	(2, 1)	(3, 2)	(1, 1)	(4, 4)
7	(6, 1)	(4, 2)	(3, 2)	(2, 1)	(4, 4)	(1, 1)

Every diagonal entry is (1, 1), the bite at (2, 2), in accordance with Theorem 4.4.

D.7 The cone eigenvalue

Convergence of the method of particular solutions for the $(\pi/3, \pi/2, 2\pi/3)$ triangle of Section 8, as the number n_b of basis functions increases; σ is the Betcke–Trefethen subspace angle at the minimum:

n_b	ν	σ
8	3.2409029943555	1.11×10^{-6}
10	3.2409029943607	9.16×10^{-8}
12	3.2409029943607	8.42×10^{-9}
14	3.2409029943607	8.33×10^{-10}

Calibration on the A_3 chamber, whose exact value is $\nu = 6$, gives 5.40×10^{-10} error at the same settings. The Rayleigh–Ritz upper bound, computed on a twelve-dimensional admissible trial space, is $\lambda_1 \leq 13.7540986066788$, i.e. $\nu \leq 3.24220504604956$, with the second Ritz value 26.0257.

D.8 Geode values

$G[1000^{(D)}]$ was computed for $D = 4, \dots, 10$. The five-dimensional value, which is the object of the challenge, has 8367 decimal digits. The four-dimensional value serves as the acceptance test and agrees with Rubine’s published digits at both ends. The full decimal expansions are in the repository of Appendix D; reproducing them here would occupy several pages to no purpose.

D Repository index

All material accompanying this paper is consolidated in a single repository,

github.com/jaideepsaipadhi/zeilberger-challenges

which supersedes the five separate repositories in which the work was originally released. Its layout follows the parts of this paper:

Directory	Contents
<code>paper/</code>	this paper, source and compiled
<code>chomp/</code>	§4: three solvers, track dumps, bite tables
<code>restricted-permutations/</code>	§5: note and four verification scripts
<code>solid-syt/</code>	§7: paper, derivation trail, operator check
<code>geode/</code>	§10: note, CRT driver, computed values
<code>kz-constants/</code>	§9: paper, verification suite, constant determination
<code>a22-partial/</code>	§6: the reduction and its obstruction
<code>cone-exponent/</code>	§8: eigenvalue determination and bounds

Each directory carries the complete contents of the corresponding original repository — engines, logs, data and papers — together with the verification scripts written for this paper. Every quantitative claim made here is reproducible from the code and data there, and each script states in its header what it checks and against what.

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